Gowin Customized PHY IP

User Guide

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Revision History

Date	Version	Description
07/21/2023	1.0E	Initial version published.
09/08/2023	1.1E	Analog Front End (AFE) configuration option added.
10/12/2023	1.2E	● The range of TX Line Rate updated to 0.025Gbps~12.5Gbps. ● TX Line Rate Ratio option and its descriptions added.
03/07/2024	1.3E	Descriptions of AFE configuration option added.
04/12/2024	1.4E	Descriptions of DRP added.
05/17/2024	1.5E	Descriptions of Clock Schematic and Reconfiguration added.
11/01/2024	1.6E	GW5AT-60 devices supported and related descriptions added.
11/08/2024	1.6.1E	Descriptions of 3.8.1 10GBASER with FIFO Mode updated.
01/17/2025	1.7E	GW5AT-15 devices supported and related descriptions added.
05/30/2025	1.7.1E	Descriptions of 3.1.3 GW5AT-15 updated.
06/17/2026	1.7.2E	Added the descriptions of the "Global Reset Enable" option in 3.2 Reference Clock.

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1 About This Guide

1.1 Purpose

Gowin Customized PHY IP User Guide is to help you learn the features and usage of Gowin Customized PHY IP by providing descriptions of features, functions, GUI, and reference design, etc. The software screenshots in this manual are based on 1.9.11. As the software is subject to change without notice, some information may not remain relevant and may need to be adjusted according to the software that is in use.

1.2 Related Documents

You can find the related documents at www.gowinsemi.com:

- SUG100, Gowin Software User Guide
- DS981, GW5AT series of FPGA Products Data Sheet
- DS1104, GW5AST series of FPGA Products Data Sheet

1.3 Terminology and Abbreviations

Table 1-1 shows the abbreviations and terminology used in this manual.

Table 1-1 Terminology and Abbreviations

Terminology and Abbreviations	Meaning
AFE	Analog Front End
ATT	Attenuator
FFE	Feed-Forward Equalization
FPGA	Field Programmable Gate Array
IP	Intellectual Property
PCS	Physical Coding Sublayer
CDR	Clock and Data Recovery
RXEQ	Receive Equalization

1.4 Support and Feedback

Gowin Semiconductor provides customers with comprehensive technical support. If you have any questions, comments, or suggestions, please feel free to contact us directly using the information provided below.

Website: www.gowinsemi.com

E-mail: support@gowinsemi.com

2 Function

2.1 Overview

Gowin Customized PHY IP supports flexible configuration of Gowin SerDes functions such as Line Rate, Reference Clock, Data Width, 8B10B Encoding/Decoding, Channel Bonding, and RX Clock Correction, etc. The GW5AT-60/15 devices not only have the above functions, but also support 64B66B and 64B67B functions.

Table 2-1 Gowin Customized PHY IP Overview

Gowin Customized IP	
Logic Resource	See 2.3 Resource Utilization
Delivered Doc.	
Design Files	Verilog (encrypted)
Reference Design	Verilog
TestBench	Verilog
Test and Design Flow	
Synthesis Software	GowinSynthesis
Application Software	Gowin Software (V1.9.9 Beta-1 and above)

Note!

For the devices supported, you can click here to get the information.

2.2 Features

- Supports line rate configuration with the range of 1Gbps~12.5Gbps
- Supports low TX line rate, with a minimum of 25 Mbps
- Supports reference clock frequency configuration with the range of 50MHz~800MHz
- Supports PLL with options of CPLL or QPLL
- Supports user data widths of 8/10/16/20/32/40/64/80
- Supports 8B10B encoding and decoding
- Supports Word Alignment
- Supports Channel Bonding

- Supports RX Clock Correction
- GW5AT-60/15 supports 64B66B and 64B67B functions
- GW5AT-60/15 supports RX Only and TX Only functions

2.3 Resource Utilization

Gowin Customized PHY IP is only used to configure SerDes and occupies almost no Fabric resources.

3 Functional Description

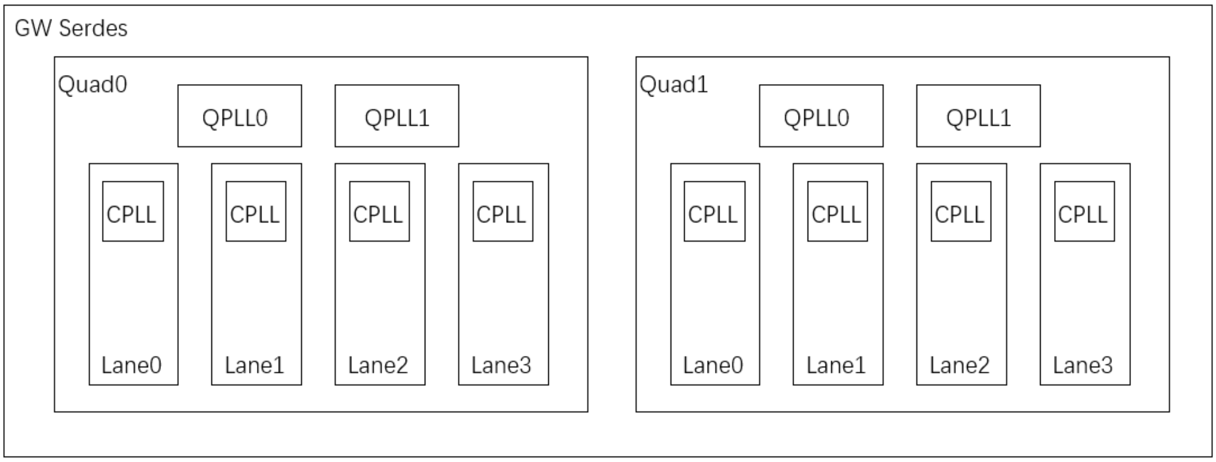
3.1 System Block Diagram

3.1.1 GW5AT-138

Gowin SerDes includes 2 Quads, and each Quad contains 4 separate lanes; there are a total of 8 lanes available to users. Each Quad contains 2 QPLLs and 4 CPLLs, where the QPLL can be shared by the 4 lanes of the Quad where it is located, and the CPLL can only be used by the lane where it is located.

Each Quad has two independent reference clock input pins, and the input reference clock can be used as the reference clock source for QPLL and CPLL. The SerDes block diagram is shown in Figure 3-1.

Figure 3-1 SerDes Block Diagram for GW5AT-138

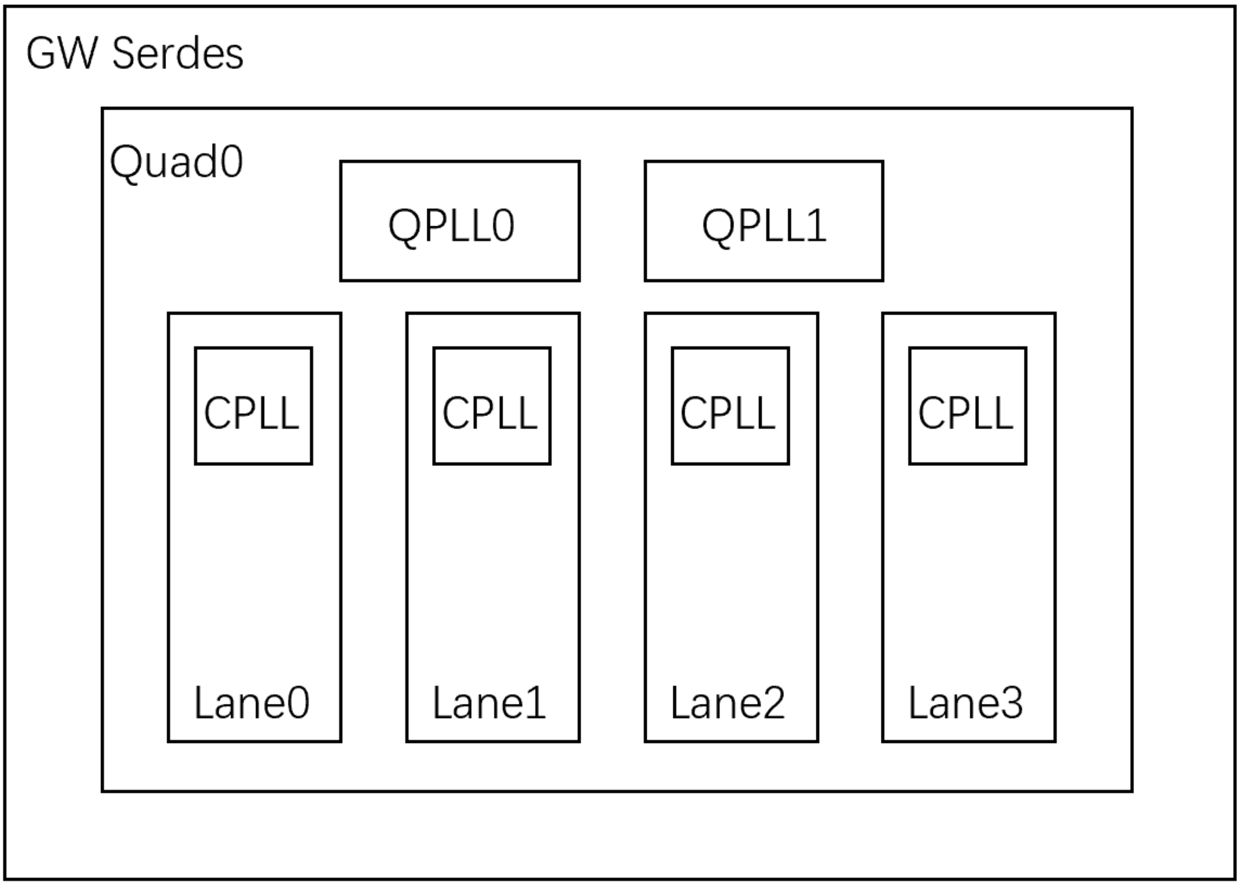


3.1.2 GW5AT-60

Gowin SerDes includes one Quad, and the Quad contains 4 separate lanes available to users. The Quad contains 2 QPLLs and 4 CPLLs, where the QPLL can be shared by the 4 lanes of the Quad, and the CPLL can only be used by the lane where it is located.

GW5AT-60 has 6 independent reference clock input pins, four of which are SerDes-dedicated clock input pins, and two of which are multiplexed with GPIOs. The input reference clock can be served as the reference clock source for the QPLL and CPLL. The selection of the reference clock needs to be determined based on the package of the selected chip. The SerDes block diagram is shown in Figure 3-2.

Figure 3-2 SerDes Block Diagram for GW5AT-60



3.1.3 GW5AT-15

The block diagram of the GW5AT-15 SerDes system is the same as that of the GW5AT-60, as shown in Figure 3-2. The difference is that GW5AT-15 has 3 independent reference clock input pins and 1 internal OSC clock input. Among the independent reference clock input pins, 2 are SerDes-dedicated clock input pins, and 1 is multiplexed with GPIO. The input reference clocks can serve as the reference clock sources for the QPLL and CPLL. The selection of the reference clock needs to be determined based on the package of the chosen chip.

3.2 Reference Clock

The proper operation of SerDes requires a stable reference clock. Users need to ensure the correctness and stability of the reference clock frequency during the operation of SerDes.

The SerDes typically requires its reference clocks to be stable before FPGA configuration completes. If the reference clocks become stable only after FPGA configuration in the target application, perform the following steps:

1. Enable the "Global Reset Enable" option in the SerDes IP configuration interface.
2. When this option is enabled, the generated SerDes IP includes a por_n_i input port.
3. After FPGA configuration is complete, set por_n_i to 0.
4. The user is required to determine whether all SerDes reference clocks are stable.
5. Once all reference clocks are stable, set por_n_i to 1.

3.2.1 GW5AT-138

Overview

GW5AT-138 has a total of 4 reference clock inputs, with Quad0 and Quad1 each containing 2 reference clocks. Reference clocks are allowed to cross Quad. In the same configuration, only one reference clock is allowed to cross Quad. Table 3-1 lists the correspondence between the schematic package and reference clocks on the IP interface.

Table 3-1 Reference Clock Correspondence for GW5AT-138

Reference Clock	Schematic Package	IP Option
Quad0 Refclk0	Q0_REFCLKP/M_0	Q0 REFCLK0
Quad0 Refclk1	Q0_REFCLKP/M_1	Q0 REFCLK1
Quad1 Refclk0	Q1_REFCLKP/M_0	Q1 REFCLK0
Quad1 Refclk1	Q1_REFCLKP/M_1	Q1 REFCLK1

Configuration

Users can configure the reference clocks through the IP interface. Reference clock configuration includes Auto and Manual modes.

When the user selects Auto mode, you need to select "Auto" Mode in the "Refclk Selection" option on the interface. In this mode, only "Reference Clock Source", "Reference Clock Frequency", and "PLL Selection" can be configured. The configuration steps are as follows:

1. Reference Clock Source: Used to select the source of the reference clock.
2. Reference Clock Frequency: Used to set the frequency of the reference clock source selected in Step 1.
3. PLL Selection: Used to select the PLL to be used.
4. Click "Calculate". If the above configuration is correct, a "Success" dialog box will pop up.

When the user selects Manual mode, you need to select "Manual" Mode in the "Refclk Selection" option on the interface. In this mode, only "PLL Selection", "REFMUX0/1", "PLL Refclk Source", and "RX Refclk Source" can be configured. The configuration steps are as follows:

1. PLL Selection: Used to select the PLL to be used.

2. PLL Refclk Source: Used to choose whether the PLL clock source comes from REFMUX0 or REFMUX1.
3. If you select REFMUX0 in Step 2, then select REFMUX0 in REFMUX0 Source option and set the clock frequency of this reference clock. If you select REFMUX1 in Step 2, then select REFMUX1 in REFMUX1 Source option and set the clock frequency of this reference clock.
4. RX Refclk Source: Used to select whether the RX reference clock source comes from REFMUX0 or REFMUX1.
5. If you select REFMUX0 in Step 4, then select REFMUX0 in REFMUX0 Source option and set the clock frequency of this reference clock. If you select REFMUX1 in Step 4, then select REFMUX1 in REFMUX1 Source option and set the clock frequency of this reference clock.
6. If you need to configure the same REFMUX in Step 3 and Step 5, the REFMUX option needs to be configured only once.
7. Click "Calculate". If the above configuration is correct, a "Success" dialog box will pop up.

3.2.2 GW5AT-60

Overview

GW5AT-60 has 6 independent reference clock input pins, four of which are SerDes-dedicated clock input pins, and two of which are multiplexed with GPIOs. Table 3-2 lists the correspondence between the schematic package and reference clocks on the IP interface.

Table 3-2 Reference Clock Correspondence for GW5AT-60

Reference Clock	Schematic Package	IP Option	IP Pin
Quad0 Refclk0	Q0_REFCLKP/M_0	Q0 REFCLK0	-
Quad0 Refclk1	Q0_REFCLKP/M_1	Q0 REFCLK1	-
Quad0 Refclk2	Q0_REFCLKP/M_2	Q0 REFCLK2	-
Quad0 Refclk3	Q0_REFCLKP/M_3	Q0 REFCLK3	-
GPIO0	Q0REF_T/C_IN0	Q0 REFIN0	gpio_refclk0_i
GPIO1	Q0REF_T/C_IN1	Q0 REFIN1	gpio_refclk1_i

If the user wants to use GPIO0 or GPIO1 as the SerDes reference clock, the user needs to connect the IP's gpio_refclk0_i or gpio_refclk1_i to the top-level input when instantiating the SerDes IP, and constrain the corresponding IO positions in the .cst file. If it is not used, the gpio_refclk0_i or gpio_refclk1_i pins can be floated

Configuration

Users can configure the reference clocks through the IP. Reference clock configuration includes Auto and Manual modes.

When the user selects Auto mode, you need to select "Auto" Mode in the "Refclk Selection" option on the interface. In this mode, only

"Reference Clock Source", "Reference Clock Frequency", and "PLL Selection" can be configured. The configuration steps are as follows:

1. Reference Clock Source: Used to select the source of the reference clock.
2. Reference Clock Frequency: Used to set the frequency of the reference clock source selected in Step 1.
3. PLL Selection: Used to select the PLL to be used.
4. Click "Calculate". If the above configuration is correct, a "Success" dialog box will pop up.

When the user selects Manual mode, you need to select "Manual" Mode in the "Refclk Selection" option on the interface. In this mode, the configuration steps are as follows:

1. PLL Selection: Used to select the PLL to be used.
2. PLL Refclk Source: Used to choose whether the PLL clock source comes from REFMUX0, REFMUX1, REFMUX2, or REFMUX3.
3. If you select REFMUX0 in Step 2, then select REFMUX0 in REFMUX0 Source option and set the clock frequency of this reference clock. If you select REFMUX1 in Step 2, then select REFMUX1 in REFMUX1 Source option and set the clock frequency of this reference clock. If you select REFMUX2 in Step 2, then select REFMUX2 in REFMUX2 Source option and set the clock frequency of this reference clock. If you select REFMUX3 in Step 2, then select REFMUX3 in REFMUX3 Source option and set the clock frequency of this reference clock.
4. RX Refclk Source: Used to select whether the RX reference clock source comes from REFMUX0, REFMUX1, REFMUX2, or REFMUX3.
5. If you select REFMUX0 in Step 4, then select REFMUX0 in REFMUX0 Source option and set the clock frequency of this reference clock. If you select REFMUX1 in Step 4, then select REFMUX1 in REFMUX1 Source option and set the clock frequency of this reference clock. If you select REFMUX2 in Step 4, then select REFMUX2 in REFMUX2 Source option and set the clock frequency of this reference clock. If you select REFMUX3 in Step 4, then select REFMUX3 in REFMUX3 Source option and set the clock frequency of this reference clock.
6. If you need to configure the same REFMUX in Step 3 and Step 5, the REFMUX option needs to be configured only once.
7. Click "Calculate". If the above configuration is correct, a "Success" dialog box will pop up.

3.2.3 GW5AT-15

Overview

GW5AT-15 SerDes has 3 independent reference clock input pins and 1 internal OSC clock input. Table 3-3 lists the correspondence between the schematic package and reference clocks on the IP interface.

Table 3-3 Reference Clock Correspondence for GW5AT-15

Reference Clock	Schematic Package	IP Option	IP Pin
Quad0 Refclk0	Q0_REFCLKP/M_0	Q0 REFCLK0	-
Quad0 Refclk1	Q0_REFCLKP/M_1	Q0 REFCLK1	-
GPIO	Q0REF_T/C_IN	Q0 REFIN	gpio_refclk_i
Internal Oscillator (OSC)	-	MCLK	mclk_i

If the user wants to use GPIO as the SerDes reference clock, he needs to connect the IP's gpio_refclk_i to the top-level input when instantiating the SerDes IP, and constrain the corresponding IO positions in the .cst file. If it is not used, the gpio_refclk_i pin can be left floated.

If the user wants to use the FPGA's internal OSC as the reference clock, he needs to instantiate the OSCB primitive in the design and connect the OSCREF of the OSCB primitive to the mclk_i of the IP. If it is not used, the mclk_i pin can be floated.

Configuration

Users can configure the reference clocks through the IP. Reference clock configuration includes Auto and Manual modes.

When the user selects Auto mode, you need to select "Auto" Mode in the "Refclk Selection" option on the interface. In this mode, only "Reference Clock Source", "Reference Clock Frequency", and "PLL Selection" can be configured. The configuration steps are as follows:

1. Reference Clock Source: Used to select the source of the reference clock.
2. Reference Clock Frequency: Used to set the frequency of the reference clock source selected in Step 1.
3. PLL Selection: Used to select the PLL to be used.
4. Click "Calculate". If the above configuration is correct, a "Success" dialog box will pop up.

When the user selects Manual mode, you need to select "Manual" Mode in the "Refclk Selection" option on the interface. In this mode, the configuration steps are as follows:

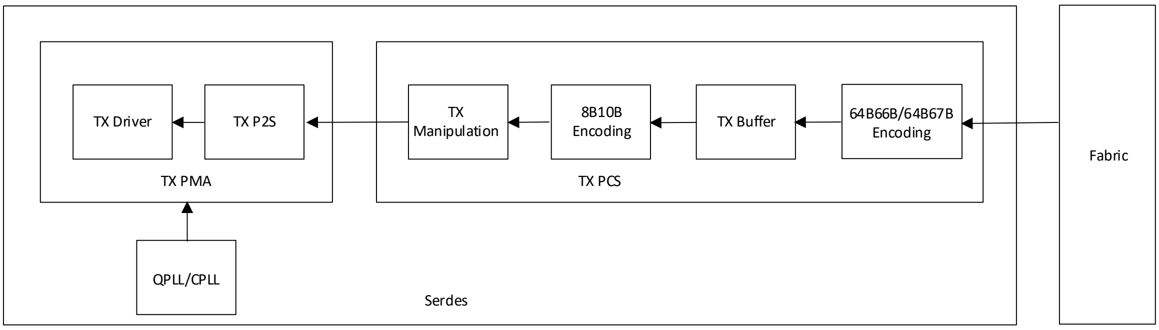
1. PLL Selection: Used to select the PLL to be used.
2. PLL Refclk Source: Used to choose whether the PLL clock source comes from REFMUX0, REFMUX1, GPIO, or MCLK.
3. If you select REFMUX0 in Step 2, then select REFMUX0 in REFMUX0 Source option and set the clock frequency of this reference clock. If you select REFMUX1 in Step 2, then select REFMUX1 in REFMUX1 Source option and set the clock frequency of this reference clock. If you select GPIO in Step 2, then set the clock frequency of GPIO. If you select MCLK in Step 2, then set the clock frequency of MCLK.

4. Click "Calculate". If the above configuration is correct, a "Success" dialog box will pop up.

3.3 Modules

3.3.1 Transmit

Figure 3-3 SerDes Transmit

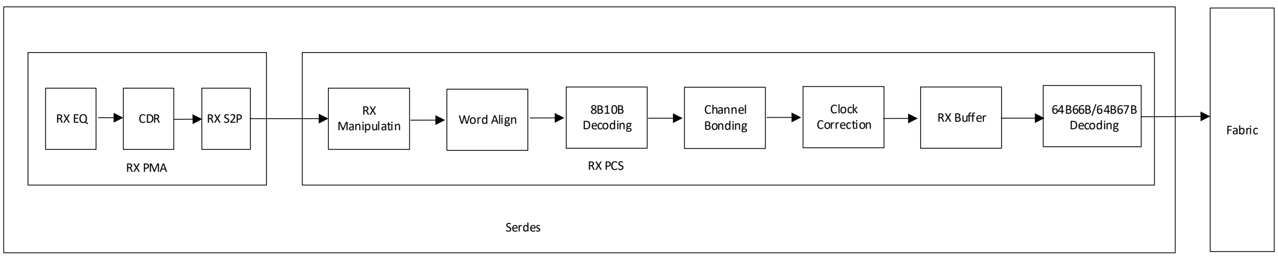


As shown above, the transmit consists of the following six modules, of which the 64B66B/64B67B Encoding module is only supported by GW5AT-60, and the modules in the TX PCS can be bypassed independently.

- TX Driver: Provides the driver for the analog of the transmit lane to convert the serial data to differential data to output to the IO of chip.
- TX P2S: Converts TX parallel data of PCS to serial data and outputs to TX Driver, and the parallel data supports 8/10/16/20 bit widths.
- TX Manipulation: Before the data is transmitted to the TX P2S, parallel data can be manipulated such as bit polarity inversion, high and low bit swapping, and byte inversion; and this module supports bypass.
- 8B10B Encoding: Realizes 8B10B encoding, and this module supports bypass.
- TX Buffer: Realizes the TX data width conversion, and supports 1:1, 1:2, 1:4 ratios. At the same time, the frequency of the clock output to the Fabric is reduced according to the ratio.
- 64B66B/64B67B Encoding: Implements 64B66B and 64B67B encoding functions.

3.3.2 Receive

Figure 3-4 SerDes Receive



As shown above, the receive consists of the following ten modules, of which the 64B66B/64B67B Encoding module is only supported by GW5AT-60, and the modules in the TX PCS can be bypassed independently.

- RXEQ: Equalizes the RX differential data to provide a stable eye diagram for CDR module.
- CDR: Extracts the clock of RX data and aligns the data center position.
- S2P: Converts the RX serial data to parallel data and outputs to PCS, and the parallel data supports 8/10/16/20 bit widths.
- RX Manipulation: After the data enters PCS, parallel data can be manipulated such as bit polarity inversion, high and low bit swapping, byte inversion, and this module supports bypass.
- Word Align: Slips auto-received data to achieve the function of aligning K character, and this module supports bypass.
- 8B10B Decoding: 8B10B decodes RX data, and this module supports bypass.
- Channel Bonding: Aligns RX data of multiple channels, and this module supports bypass.
- Clock Correction: Realizes clock correction for RX data by adding or removing certain fields, and this module supports bypass.
- RX Buffer: Realizes RX data width conversion, and supports 1:1, 1:2, 1:4 ratios; at the same time, the frequency of the clock output to the Fabric is reduced according to the ratio.
- 64B66B/64B67B Encoding: Implements 64B66B and 64B67B encoding functions.

3.4 Operation Mode

Users can configure the operating mode of the lane. The operating modes include TX and RX, TX Only, and RX Only.

When both the TX and RX of a lane are used simultaneously, the operating mode should be configured as TX and RX. In this case, both the receive and transmit data of the lane are valid.

When only the TX of a lane is used, the operating mode can be configured as TX Only. In this case, only the TX lane is enabled, and the RX lane is disabled, which helps reduce power consumption.

When only the RX of a lane is used, the operating mode can be configured as RX Only. In this case, only the RX lane is enabled, and the TX lane is disabled, which helps reduce power consumption.

3.5 User Clock

3.5.1 Transmit Clock

The transmit clock is generated by the CPLL/QPLL. When configuring the TX lane, you need to configure the TX lane rate, the PLL used

(CPLL/QPLL), and the reference clock source and its frequency. Based on the above configuration, the IP configures SerDes PLL to generate a high-speed clock for data transmission. At the same time, SerDes will divide the high-speed clock according to the user configuration; then output the clock to Fabric to be used as Fabric TX clock. The frequency of the TX clock outputted to Fabric is calculated as follows:

- If the TX lane rate is ≥ 1Gbps, then

  F=TX lane rate[1]/Fabric data width

Note!

[1]The input value of TX Line Rate on the GUI.

For example, if the user configures the TX data rate to be 1.25Gbps and configures the TX parallel data width to be 40 bits, the Fabric TX clock is 1.25Gbps/40=31.25MHz.

- If the TX lane rate is < 1Gbps, then

  F=TX lane rate[1] X Ratio[2]/Fabric data width

Note!

- [1] The input value of TX Line Rate on the GUI.
- [2] The Ratio value on the GUI with the options 5x=5, 10x=10, 20x=20, 40x=40

For example, if the user configures TX data rate to be 0.5Gbps, the Ratio is configured to be 5x, and the TX parallel data width is configured to be 40 bits, then the Fabric TX clock will be (0.5Gbps×5)/40=62.5MHz.

3.5.2 Receive Clock

When configuring the RX lane, you need to configure the RX lane rate.

The RX clock is recovered from data through CDR, and the recovered serial data clock output from CDR is used for the RX S2P module. At the same time, SerDes will divide the serial data clock according to the user configuration; then output the clock to Fabric to be used as Fabric RX clock. The frequency of the RX clock outputted to Fabric is calculated as follows:

F= RX lane rate/Fabric data width

For example, if the user configures the RX data rate to be 1.25Gbps and configures the RX parallel data width to be 40 bits, the Fabric TX clock is 1.25Gbps/40=31.25MHz.

3.6 User Data

3.6.1 Transmit Data

TX Data Width

TX Data Width is determined by the Internal Data Width and TX External Data Ratio options.

You can configure SerDes internal data width through Internal Data Width, and configure the ratio of TX Buffer converting the internal width to the user interface width through TX External Data Ratio. When the user selects a larger width, the priority is to configure the Internal Data Width to the maximum. For example, when you configure a bit width of 40, you need to configure Internal Data Width=20 and TX External Data Ratio=1:2.

Note!

- For GW5AT-138, the Internal Data Width for both receive and transmit directions must be the same.
- For GW5AT-60, the Internal Data Width for receive and transmit directions can be configured independently.

8B10B Encoding

8B10B Encoding module can realize TX Data 8B10B encoding. When the Internal Data Width is configured to 10 or 20, the user can choose to enable/disable this function. When the Internal Data Width is configured to 8 or 16, the user cannot enable this function.

When the user disables this function, the data transmitted from the Fabric is Raw Data; when the user enables this function, the data transmitted from the Fabric is the K character indication and the data before encoding.

Channel Bonding

The Channel Bonding module can realize the bonding of multiple TX channels. When the user does not check the option, this function is disabled. When the user checks the option, SerDes uses TX Buffer to achieve this function. After enabling this function, you need to configure Master Channel; in general, you can choose any one in the selected channel. Finally, you need to configure Read Start Depth to control read start depth of the TX Buffer.

3.6.2 Receive Data

RX Data Width

RX Data Width is determined by the Internal Data Width and RX External Data Ratio options.

You can configure SerDes internal data width through Internal Data Width, and configure the ratio of RX Buffer converting the internal width to the user interface width through RX External Data Ratio. For example, when you configure a bit width of 40, you need to configure Internal Data Width=20 and RX External Data Ratio=1:2.

Note!

- For GW5AT-138, the Internal Data Width for both receive and transmit directions must be the same.
- For GW5AT-60, the Internal Data Width for receive and transmit directions can be configured independently.

Word Alignment

The Word Alignment module can realize RX data alignment. When the user enables this function, the module aligns the boundary of the RX parallel data according to the Pattern configured by the user, so that the output parallel data is consistent with the configured K character. When the user enables 8B10B decoding function, this module needs to be enabled.

8B10B Decoding

8B10B Encoding module can realize RX Data 8B10B decoding. When the Internal Data Width is configured to 10 or 20, the user can choose to enable/disable this function. When the Internal Data Width is configured to 8 or 16, the user cannot enable this function.

When the user disables this function, the data received from the Fabric is Raw Data; when the user enables this function, the data received from the Fabric is the K character indication and the data after decoding.

Channel Bonding

The Channel Bonding module can realize the bonding of multiple RX channels. When the user selects None, this function is disabled. When the user selects One Word/Two Words/Four Words, IP automatically enables this function, and the number of aligned Pattern is configured. After enabling this function, you need to configure Master Channel; in general, you can choose any one in the selected channel. Finally, align Pattern, Max Skew, and Read Start Depth. Pattern and Max Skew alignment is configured according to the specific protocol. Pattern alignment supports a maximum of 4 bytes, and the first Pattern must be a K character.

When the user enables this function, 8B10B Decoding and Word Alignment must be enabled first.

Clock Correction

The Clock Correction module can realize the conversion of RX data clock domain. When the user selects None, this function is disabled. When the user selects One Word/Two Words, IP automatically enables this function, and the number of Correction Pattern is configured. After enabling this function, you need to configure Master Channel; in general, you can choose any one in the selected channel. Finally, configure Correction Pattern and Read Start Depth. Correction Pattern is configured according to the specific protocol. Correction Pattern supports a maximum of 2 bytes, and the first Pattern must be a K character.

The Clock Source option in RX Clock Correction page allows the user to set the RX clock source, i.e., which clock source to place the RX clock to. User option means setting Clock Source as Fabric input clock (e.g. q0_ln0_cc_clk_i) and TX option means setting Clock Source as TX clock. The q0_ln0_cc_clk_i input clock frequency is:

- For GW5AT-138, F=RX Lane Rate/Internal Data Width
- For GW5AT-60, F=RX Lane Rate/RX Internal Data Width

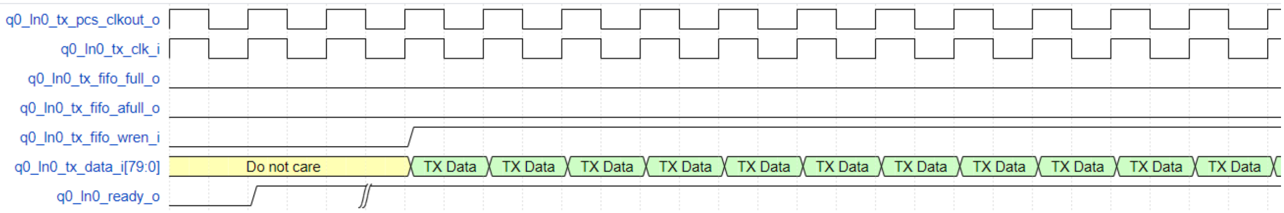
3.7 User Interface (No Encoding or 8B10B Encoding/Decoding)

SerDes user interface is in the form of FIFO, and its read and write operations are similar to those of FIFO.

3.7.1 Transmit Data

TX Line Rate≥1Gbps

Figure 3-5 Transmit Data Interface Timing When TX Rate≥1Gbps



As shown in Figure 3-5, Quad0 Lane0 is used as an example.

- q0_ln0_tx_pcs_clkout_o is the SerDes PCS TX clock, which is used as the read clock for the TX Buffer.
- q0_ln0_tx_clk_i is the input clock, which is used as the write clock for TX Buffer. This clock can be directly connected to q0_ln0_tx_pcs_clkout_o to achieve the same frequency of TX Buffer read and write clocks.
- q0_ln0_tx_fifo_full_o is the TX Buffer full signal; 1 means full and 0 means non-full.
- q0_ln0_tx_fifo_afull_o is the TX Buffer almost full signal; 1 means almost full and 0 means non-almost full.
- q0_ln0_tx_fifo_wren_i is TX Buffer write enable; 1 means write valid and 0 means write invalid.
- q0_ln0_tx_data_i is the write data for TX Buffer. When q0_ln0_tx_fifo_wren_i is 1, q0_ln0_tx_data_i writes TX Buffer. When q0_ln0_tx_fifo_wren_i is 0, q0_ln0_tx_data_i does not write TX Buffer.
- q0_ln0_tx_fifo_wrusewd_o indicates how much data is stored in the TX Buffer.

As shown in the Figure 3-5, when TX Buffer read and write clocks are at the same frequency, q0_ln0_tx_fifo_full_o and q0_ln0_tx_fifo_afull_o are constant 0, indicating that TX Buffer always has space left for data write. At this point, you can set q0_ln0_tx_fifo_wren_i to 1 and q0_ln0_tx_data_i is written as a continuous data stream.

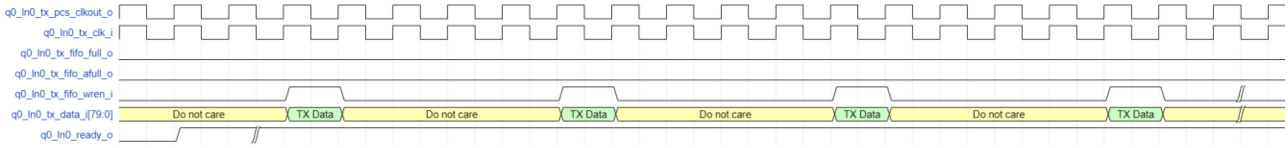
TX Line Rate < 1Gbps

When TX Line Rate < 1Gbps is configured, the corresponding Ratio configuration option on the interface is enabled. The user needs to ensure that TX Line Rate X Ratio[1] ≥ 1Gbps.

Note!

[1] The Ratio value on the GUI with the options 5x=5, 10x=10, 20x=20, 40x=40

Figure 3-6 Transmit Data Interface Timing When TX Rate<1Gbps



As shown in Figure 3-6, using Quad0 Lane0, Ratio=5x as an example:

- q0_ln0_tx_pcs_clkout_o is the SerDes PCS TX clock, which is used as the read clock for TX Buffer.
- q0_ln0_tx_clk_i is the input clock, which is used as the write clock for TX Buffer. This clock can be directly connected to q0_ln0_tx_pcs_clkout_o to achieve the same frequency of TX Buffer read and write clocks.
- q0_ln0_tx_fifo_full_o is the TX Buffer full signal; 1 means full and 0 means non-full.
- q0_ln0_tx_fifo_full_o is the TX Buffer almost full signal; 1 means almost full and 0 means non-almost full.
- q0_ln0_tx_fifo_wren_i is the TX Buffer write enable, 1 means write valid and 0 means write invalid.
- q0_ln0_tx_data_i is the write data for TX Buffer. When q0_ln0_tx_fifo_wren_i is 1, q0_ln0_tx_data_i writes TX Buffer. When q0_ln0_tx_fifo_wren_i is 0, q0_ln0_tx _data_i does not write buffer.
- q0_ln0_tx_fifo_wrusewd_o indicates how much data is stored in the TX Buffer.

As shown in the figure, when the TX Buffer read and write clocks are at the same frequency, data can be written every 5 clock cycles, at which time q0_ln0_tx_fifo_full_o and q0_ln0_tx_fifo_afull_o are constant 0, indicating that the TX Buffer has always has space left for data write.

Similarly, when Ratio=10x, data is written every 10 clock cycles; when Ratio=20x, data is written every 20 clock cycles; when Ratio=40x, data is written every 40 clock cycles. The above operations ensure that the TX Buffer read and write operations are performed at the same cycles, and without read empty or write full.

In addition to the above operation, users can write data according to the mode of writing to FIFO according to the design requirements. For example, when the TX Buffer is detected to be full, data can be written.

q0_ln0_tx_data_i is the TX data input with 80 bits, low first. The meaning represented by each bit is different in different encoding and bit width modes, which can be referred in Table 3-4, Table 3-5, and Table 3-6.

Table 3-4 x8 Bit Width Configuration When 8B10B Encoding Disabled

q0_ln0_tx_data_i bit[n]	width=8	width=16	width=32	width=64
7~0	7~0	7~0	7~0	7~0
8	N/A	N/A	N/A	N/A
9	N/A	N/A	N/A	N/A
17~10	N/A	15~8	15~8	15~8
18	N/A	N/A	N/A	N/A
19	N/A	N/A	N/A	N/A
27~20	N/A	N/A	23~16	23~16
28	N/A	N/A	N/A	N/A
29	N/A	N/A	N/A	N/A
37~30	N/A	N/A	31~24	31~24
38	N/A	N/A	N/A	N/A
39	N/A	N/A	N/A	N/A
47~40	N/A	N/A	N/A	39~32
48	N/A	N/A	N/A	N/A
49	N/A	N/A	N/A	N/A
57~50	N/A	N/A	N/A	47~40
58	N/A	N/A	N/A	N/A
59	N/A	N/A	N/A	N/A
67~60	N/A	N/A	N/A	55~48
68	N/A	N/A	N/A	N/A
69	N/A	N/A	N/A	N/A
77~70	N/A	N/A	N/A	63~56
78	N/A	N/A	N/A	N/A
79	N/A	N/A	N/A	N/A

Note!

In a certain width mode, only the grayed bits need to be considered.

Table 3-5 x10 Bit Width Configuration When 8B10B Encoding Disabled

q0_ln0_tx_data_i bit[n]	width=10	width=20	width=40	width=80
9~0	9~0	9~0	9~0	9~0
19~10	N/A	19~10	19~10	19~10
29~20	N/A	N/A	29~20	29~20
39~30	N/A	N/A	39~30	39~30
49~40	N/A	N/A	N/A	49~40
59~50	N/A	N/A	N/A	59~50
69~60	N/A	N/A	N/A	69~60
79~70	N/A	N/A	N/A	79~70

Note!

In a certain width mode, only the grayed bits need to be considered.

Table 3-6 x10 Bit Width Configuration When 8B10B Encoding Enabled

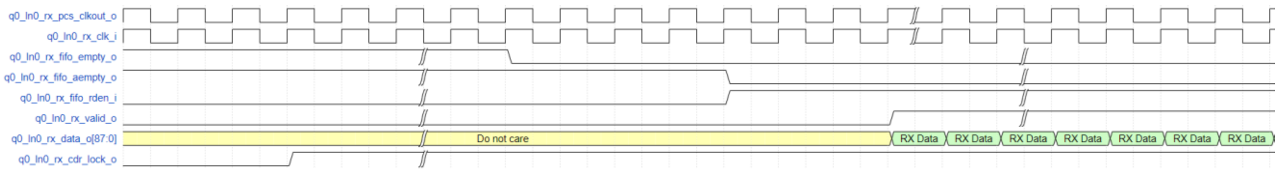
q0_ln0_tx_data_i bit[n]	width=10	width=20	width=40	width=80
7~0	Code	Code	Code	Code
8	K	K	K	K
9	N/A	N/A	N/A	N/A
17~10	N/A	Code	Code	Code
18	N/A	K	K	K
19	N/A	N/A	N/A	N/A
27~20	N/A	N/A	Code	Code
28	N/A	N/A	K	K
29	N/A	N/A	N/A	N/A
37~30	N/A	N/A	Code	Code
38	N/A	N/A	K	K
39	N/A	N/A	N/A	N/A
47~40	N/A	N/A	N/A	Code
48	N/A	N/A	N/A	K
49	N/A	N/A	N/A	N/A
57~50	N/A	N/A	N/A	Code
58	N/A	N/A	N/A	K
59	N/A	N/A	N/A	N/A
67~60	N/A	N/A	N/A	Code
68	N/A	N/A	N/A	K
69	N/A	N/A	N/A	N/A
77~70	N/A	N/A	N/A	Code
78	N/A	N/A	N/A	K
79	N/A	N/A	N/A	N/A

Note!

In a certain width mode, only the grayed bits need to be considered.

3.7.2 Receive Data

Figure 3-7 Receive Data Interface Timing



As shown in Figure 3-7, taking Quad0 Lane0 as an example, when the user monitors that q0_ln0_rx_cdr_lock_o is 1, it means that the RX CDR has entered the lock status.

- q0_ln0_rx_pcs_clkout_o is the SerDes PCS RX clock, which is used as the write clock for RX Buffer.

- q0_ln0_rx_clk_i is the input clock, which is used as the RX Buffer write clock. This clock can be directly connected to q0_ln0_rx_pcs_clkout_o to achieve the same frequency of RX Buffer read and write clocks.
- q0_ln0_rx_fifo_empty_o is the RX Buffer empty signal; 1 means empty and 0 means non-empty.
- q0_ln0_rx_fifo_aempty_o is the RX Buffer almost empty signal; 1 means almost empty and 0 means non-almost empty.
- q0_ln0_rx_fifo_rden_i is RX Buffer read enable; 1 means read valid and 0 means read invalid.
- q0_ln0_rx_data_o is the read data for RX Buffer read data. When q0_ln0_rx_fifo_rden_i is 1, read data from RX Buffer. When q0_ln0_rx_fifo_rden_i is 0, RX Buffer data is not read.
- q0_ln0_rx_valid_o indicates that q0_ln0_rx_data_o is valid. When q0_ln0_rx_fifo_rden_i is 1, the data will be output to q0_ln0_rx_data_o with a delay of 3 cycles. The user can know whether q0_ln0_rx_valid_o is valid by q0_ln0_rx_data_o. When q0_ln0_rx_valid_o is 1, q0_ln0_rx_data_o is valid. When q0_ln0_rx_valid_o is 0, q0_ln0_rx_data_o is invalid.
- q0_ln0_rx_fifo_rdusewd_o indicates how much data in the RX Buffer has not been read.

As shown in the figure, when the RX Buffer read and write clocks are in the same frequency, the user can invert q0_ln0_rx_fifo_aempty_o as q0_ln0_rx_fifo_rden_i. At this point, q0_ln0_rx_fifo_aempty_o will always be 0, and q0_ln0_rx_data_o can be read as a continuous data stream.

q0_ln0_rx_data_o is the RX data output with 88 bits, low first. The meaning represented by each bit is different in different encoding and bit width modes, which can be referred in Table 3-7, Table 3-8, and Table 3-9.

Table 3-7 x8 Bit Width Configuration When 8B10B Encoding Disabled

q0_ln0_rx_data_o bit[n]	width=8	width=16	width=32	width=64
7~0	7~0	7~0	7~0	7~0
8	N/A	N/A	N/A	N/A
9	N/A	N/A	N/A	N/A
17~10	N/A	15~8	15~8	15~8
18	N/A	N/A	N/A	N/A
19	N/A	N/A	N/A	N/A
27~20	N/A	N/A	23~16	23~16
28	N/A	N/A	N/A	N/A
29	N/A	N/A	N/A	N/A
37~30	N/A	N/A	31~24	31~24
38	N/A	N/A	N/A	N/A
39	N/A	N/A	N/A	N/A

q0_ln0_rx_data_o bit[n]	width=8	width=16	width=32	width=64
47~40	N/A	N/A	N/A	39~32
48	N/A	N/A	N/A	N/A
49	N/A	N/A	N/A	N/A
57~50	N/A	N/A	N/A	47~40
58	N/A	N/A	N/A	N/A
59	N/A	N/A	N/A	N/A
67~60	N/A	N/A	N/A	55~48
68	N/A	N/A	N/A	N/A
69	N/A	N/A	N/A	N/A
77~70	N/A	N/A	N/A	63~56
78	N/A	N/A	N/A	N/A
79	N/A	N/A	N/A	N/A
87~80	N/A	N/A	N/A	N/A

Note!

In a certain width mode, only the grayed bits need to be considered.

Table 3-8 x10 Bit Width Configuration When 8B10B Encoding Disabled

q0_ln0_rx_data_o bit[n]	width=10	width=20	width=40	width=80
9~0	9~0	9~0	9~0	9~0
19~10	N/A	19~10	19~10	19~10
29~20	N/A	N/A	29~20	29~20
39~30	N/A	N/A	39~30	39~30
49~40	N/A	N/A	N/A	49~40
59~50	N/A	N/A	N/A	59~50
69~60	N/A	N/A	N/A	69~60
79~70	N/A	N/A	N/A	79~70
87~80	N/A	N/A	N/A	N/A

Note!

In a certain width mode, only the grayed bits need to be considered.

Table 3-9 x10 Bit Width Configuration When 8B10B Encoding Enabled

q0_ln0_rx_data_o bit[n]	width=10	width=20	width=40	width=80
7~0	Code	Code	Code	Code
8	K	K	K	K
9	Disparity Error	Disparity Error	Disparity Error	Disparity Error
80	Decoder Error	Decoder Error	Decoder Error	Decoder Error
17~10	N/A	Code	Code	Code
18	N/A	K	K	K
19	N/A	Disparity Error	Disparity Error	Disparity Error

q0_ln0_rx_data_o bit[n]	width=10	width=20	width=40	width=80
81	N/A	Decoder Error	Decoder Error	Decoder Error
27~20	N/A	N/A	Code	Code
28	N/A	N/A	K	K
29	N/A	N/A	Disparity Error	Disparity Error
82	N/A	N/A	Decoder Error	Decoder Error
37~30	N/A	N/A	Code	Code
38	N/A	N/A	K	K
39	N/A	N/A	Disparity Error	Disparity Error
83	N/A	N/A	Decoder Error	Decoder Error
47~40	N/A	N/A	N/A	Code
48	N/A	N/A	N/A	K
49	N/A	N/A	N/A	Disparity Error
84	N/A	N/A	N/A	Decoder Error
57~50	N/A	N/A	N/A	Code
58	N/A	N/A	N/A	K
59	N/A	N/A	N/A	Disparity Error
85	N/A	N/A	N/A	Decoder Error
67~60	N/A	N/A	N/A	Code
68	N/A	N/A	N/A	K
69	N/A	N/A	N/A	Disparity Error
86	N/A	N/A	N/A	Decoder Error
77~70	N/A	N/A	N/A	Code
78	N/A	N/A	N/A	K
79	N/A	N/A	N/A	Disparity Error
87	N/A	N/A	N/A	Decoder Error

Note!

In a certain width mode, only the grayed bits need to be considered.

3.7.3 Status Interface

The IP provides status interfaces for users to view the channel status in real time, see details in Table 4-1.

3.8 User Interface (64B66B Encoding/Decoding)

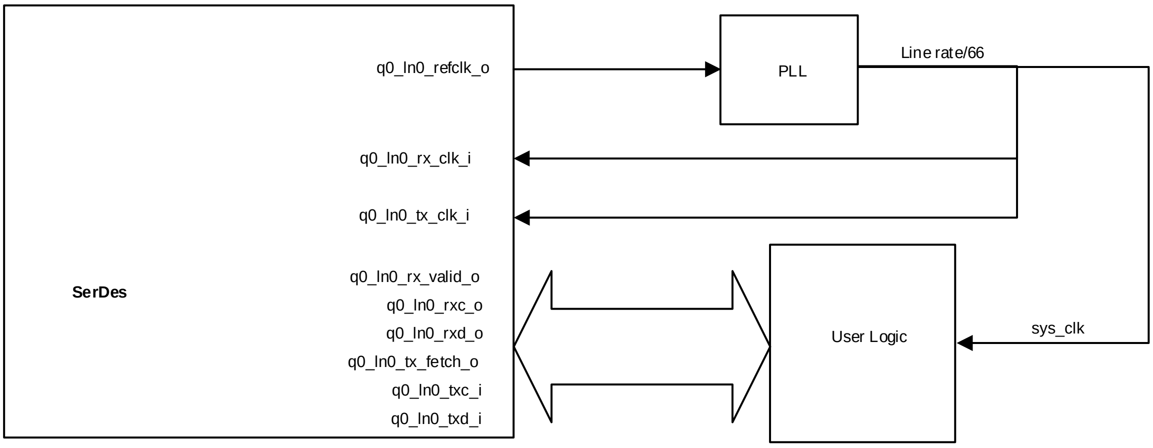
GW5AT-60 supports 64B66B encoding and decoding compliant with 10GBASE-R protocol. When using this function, users must ensure that the target protocol complies with the 10GBASE-R protocol or is a subset of 10GBASE-R protocol.

When users enable 64B66B encoding and decoding, the TX/RX Internal Data Width must be configured to 16, and the TX/RX External Data Ratio must be configured to 1:4. At this point, the encoding and decoding configuration can be set to 64B66B.

3.8.1 10GBASER with FIFO Mode

When users configure the 64B66B Mode to 10GBASER with FIFO, the CTC function is enabled on the receive side, and the receiving and transmitting clocks through the PLL can have the same source, frequency and phase.

Figure 3-8 10GBASER with FIFO Mode Clock Structure



As shown in Figure 3-8, using the q0_ln0 clock as an example, the q0_ln0_refclk_o frequency is related to the input reference clock (equal to the reference clock or the n-division frequency of the reference clock). Users can instantiate PLL in the project, and q0_ln0_refclk_o outputs Line rate/66 clock through PLL as q0_ln0_rx_clk_i and q0_ln0_tx_clk_i input clock, and can also be used as sys_clk to drive user logic. At this point, signals related to 64B66B are synchronized with sys_clk. With this clock structure, q0_ln0_tx_fetch_o and q0_ln0_rx_valid_o are always 1.

Figure 3-9 RX &TX Data Timing in 10GABSER with FIFO Mode

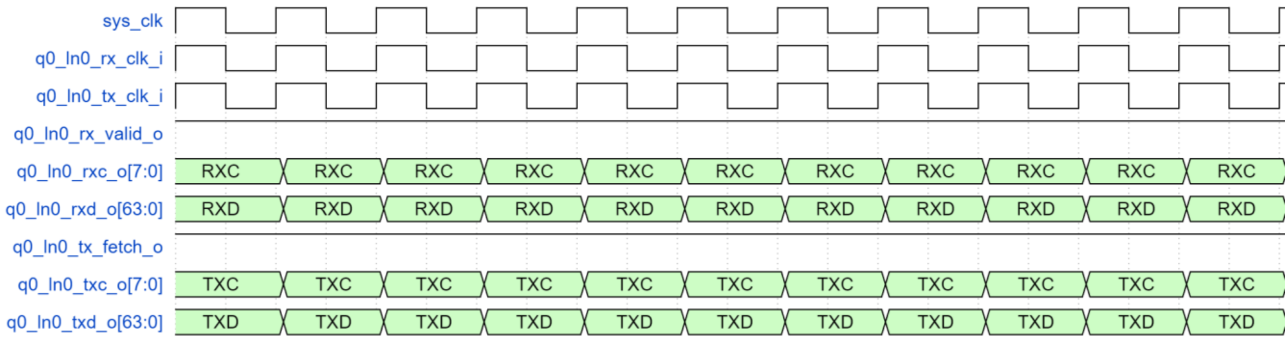
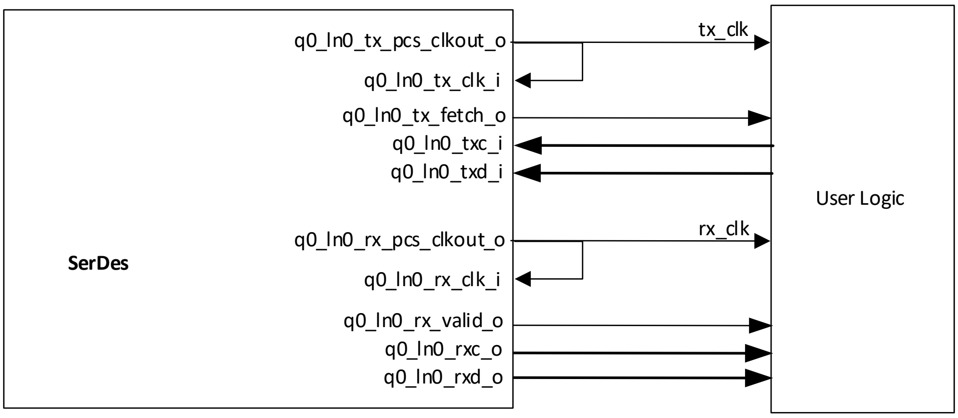


Figure 3-9 shows the timing diagram of 10GBASER with FIFO mode. q0_ln0_rxc_i, q0_ln0_rxd_i, q0_ln0_txc_o, and q0_ln0_txd_o are all synchronized with sys_clk. q0_ln0_rx_valid_o and q0_ln0_tx_fetch_o are always 1.

3.8.2 Async with FIFO Mode

When users configure the 64B66B Mode to Async with FIFO, the CTC function is disabled on the receive side, and user data is synchronized with the clock output from the SerDes for data transmission and reception. At the same time, q0_ln0_rx_valid_o and q0_ln0_tx_fetch_o will generate 1 invalid cycle every 32 valid cycles according to the clock ratio. Users need to judge these two signals to perform data TX & RX operations.

Figure 3-10 Async with FIFO Mode Clock Structure



Using q0_ln0 as an example, the clock structure is shown in Figure 3-10. The TX data is synchronized with tx_clk and the RX data is synchronized with rx_clk.

Figure 3-11 Transmit Timing in Async with FIFO Mode

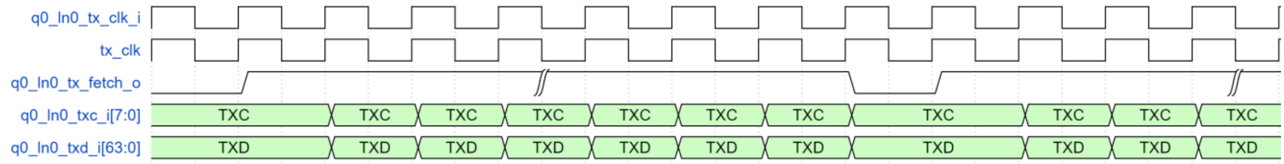


Figure 3-11 shows the transmit timing. When q0_ln0_tx_fetch_o is 1, the IP fetches TXC and TXD; when q0_ln0_tx_fetch_o is 0, the IP does not fetch TXC and TXD. Users need to ensure that all TX data is input when q0_ln0_tx_fetch_o is 1.

Figure 3-12 Receive Timing in Async with FIFO Mode

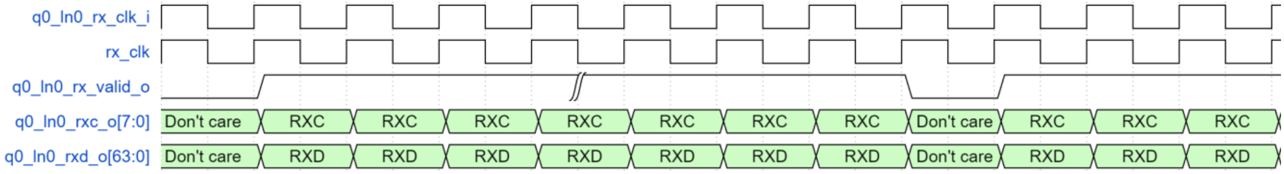
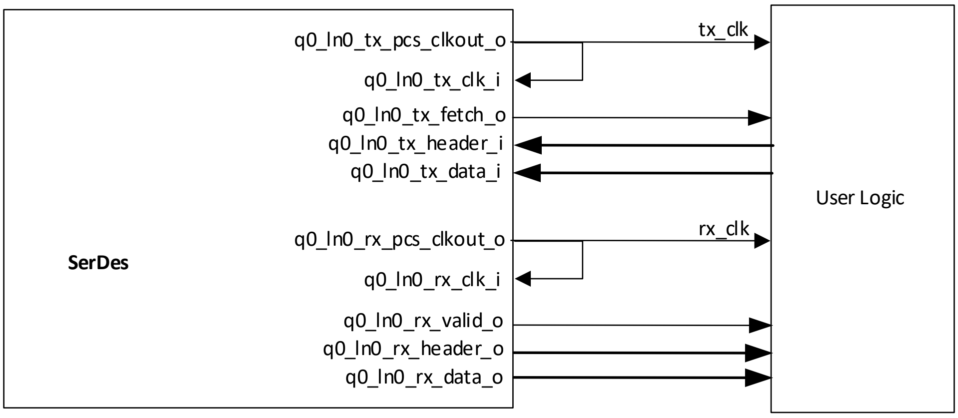


Figure 3-12 shows the receive timing. When q0_ln0_rx_valid_o is 1, RXC and RXD are valid; when q0_ln0_rx_valid_o is0, RXC and RXD are invalid.

3.9 User Interface (64B67B Encoding/Decoding)

GW5AT-60 supports 64B67B encoding and decoding compliant with the Interlaken protocol. When users configure the chip to 64B67B Mode, user data can be synchronized with the clock output from the SerDes for data transmission and reception.

Figure 3-13 64B67B Clock Structure



Using q0_ln0 as an example, the clock structure is shown in Figure 3-13. The TX data is synchronized with tx_clk and the RX data is synchronized with rx_clk.

Figure 3-14 64B67B Transmit Timing

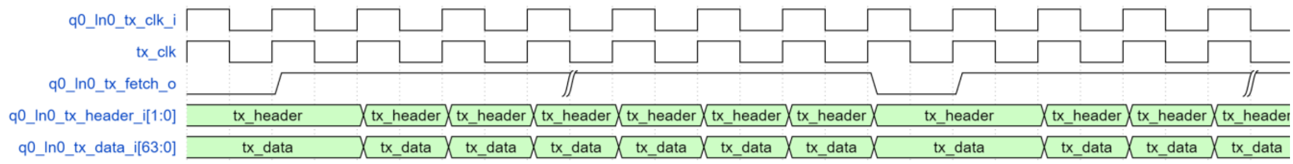


Figure 3-14 shows the transmit timing. When q0_ln0_tx_fetch_o is 1, the IP fetches tx_header and tx_data; when q0_ln0_tx_fetch_o is 0, the IP does not fetch tx_header and tx_data. Users need to ensure that all TX data is input when q0_ln0_tx_fetch_o is 1.

Figure 3-15 64B67B Receive Timing

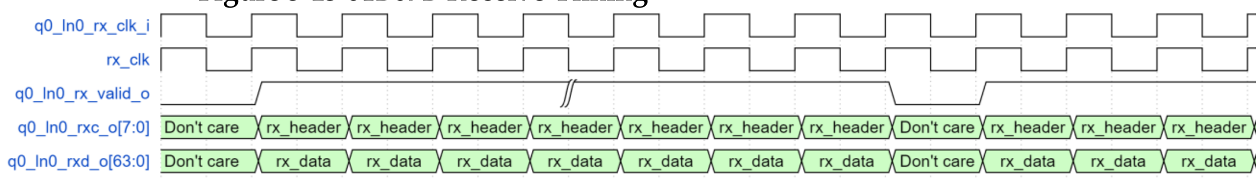


Figure 3-15 shows the receive timing. When q0_ln0_rx_valid_o is 1, rx_header and rx_data are valid; when q0_ln0_rx_valid_o is 0, rx_header and rx_data are invalid.

3.10 AFE

AFE means analog front end, and the user can configure SerDes analog parameters through the interface.

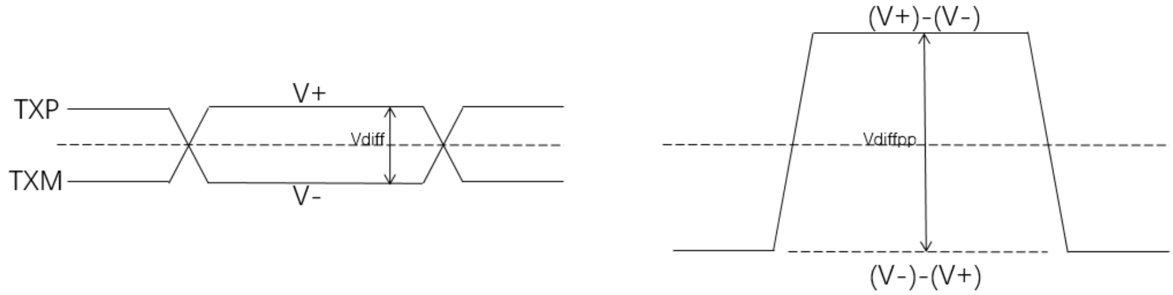
3.10.1 Transmit

On the TX side, the user can configure the differential swing and FFE parameters of the TX signals.

Transmit Differential Swing

As shown in Figure 3-16, the differential voltage of TX signal is Vdiff=(V+)-(V-), and the TX differential signal swing is Vdiffpp=2xVdiff, and the user can configure Vdiffpp through the interface with the range of 180mV~900mV.

Figure 3-16 TX Differential Signal Swing Vdiffpp



TX FFE

FFE means Feed-Forward Equalization (FFE); SerDes supports auto and manual adjustments of FFE coefficients. When the user configures FFE Mode as Auto, SerDes automatically adjusts the FFE coefficient according to the hardware environment, at this time Cm, C0 and C1 configuration is invalid. When the user configures FFE Mode as Manual, the user can manually adjust the 3-tap coefficients and configure the de-weighting of the TX signal.

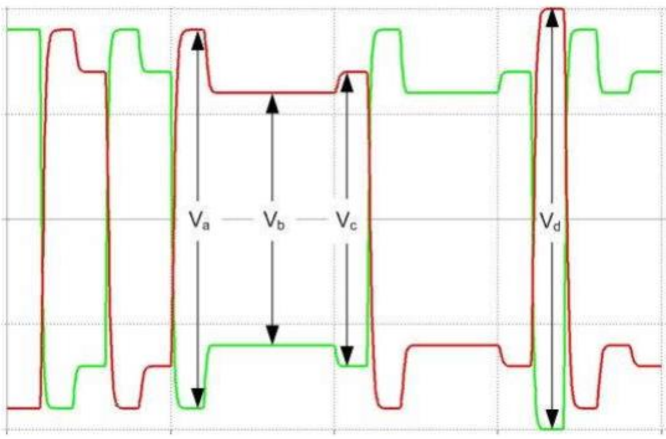
As shown in Figure 3-17, when the user configures the FFE Mode as Manual, the voltage amplitude of Va, Vb and Vc can be adjusted by adjusting the values of Cm, C0 and C1， and the calculation formula is as follows:

Va=Vdiffpp *(-Cm+C0+C1)/40

Vb=Vdiffpp *(-Cm+C0-C1)/40

Vc=Vdiffpp *(Cm+C0-C1)/40

Figure 3-17 FFE TX Voltage Definition



3.10.2 Receive

Receive Differential Signal Threshold

On the RX side, the user can configure the SD Threshold option to adjust the effective voltage threshold of RX signal. When the RX differential signal is greater than SD threshold, SerDes judges that the valid data is received; when the RX differential signal is less than SD threshold, SerDes judges that the valid data is not received, then enters the Electrical Idle state.

Receive Equalization

On the receive side, SerDes has equalization function, and users can adjust the equalizer configuration based on the data rate and channel attenuation to achieve the optimal RX status.

The equalizer in the SerDes can be divided into auto mode and manual mode. When the user configures the Equalization Mode as Auto, the equalizer operates in automatic mode. In this mode, during SerDes RX link establishment, the equalizer automatically adjusts based on the current quality of received data to achieve the optimal status. At this time, the ATT and BOOST options cannot be configured. When the user configures the Equalization Mode as Manual, the equalizer operates in manual mode. In this mode, the user needs to manually configure the ATT and BOOST options to make the equalizer to its best status.

ATT (Attenuator): Adjusts the intermediate frequency attenuation during reception. A smaller value indicates greater attenuation, with a range of 0 to 10.

BOOST (Analog Boost): Adjusts the high-frequency amplification during reception. A larger value indicates greater gain, with a range of 0 to 15.

If users configure the Equalization Mode as Manual, they need to continuously experiment with combinations of ATT and BOOST options to achieve the optimal status for SerDes reception. Therefore, it is recommended that users prioritize using the Auto mode. If the Auto mode does not adapt itself to the optimal status, users can try the Manual mode.

BIAS

The BIAS option is used to configure the amplification parameters of the SerDes for RX signals. When the RX signal rate is high and the attenuation is significant, the user can change the configuration of this option. The higher the configuration for this option, the stronger the amplification of the signal. This option is based on the QUAD configuration. When the configuration of one Lane is changed, it simultaneously changes the configuration for all Lanes in the Quad where the current Lane is located.

3.11 Dynamic Reconfiguration Function

After the initialization configuration of SerDes is completed, users can dynamically configure SerDes registers to adjust SerDes function through Dynamic Reconfiguration Port (DRP) interface. When users select the "DRP Port" in the IP interface, the IP will generate the DRP interface to implement the dynamic reconfiguration function.

Users can access the internal registers of the SerDes through the DRP interface.

During the read operation, drp_rden_i is set to high, and the read register address is input to drp_addr_i[23:0] in the first clock cycle when drp_rden_i is set to high. drp_rden_i stays high until drp_rdvld_o is set to high; the corresponding drp_rddata_o[31:0] data is the read data returned from the read operation when drp_rdvld_o is high. drp_rden_i should be set to low in the clock cycle immediately following the assertion of drp_rdvld_o to end the current read operation. The DRP interface read timing is shown in Figure 3-18.

During the write operation, drp_wren_i is set to high, and the write register address is input to drp_addr_i[23:0] in the first clock cycle when drp_wren_i is set high, and the write data is input to drp_wrdata_i[31:0] in the same clock cycle; the fixed 8 'hff is input to drp_strb_i[7:0]. drp_wren_i stays high until drp_ready_o is set to high, and drp_ready_o is set high to indicate that this write operation is complete. drp_wren_i should be set to low in the clock cycle immediately following the assertion of drp_ready_o to end the current write operation. The write timing is shown in Figure 3-19.

Figure 3-18 DRP Interface Read Timing Diagram

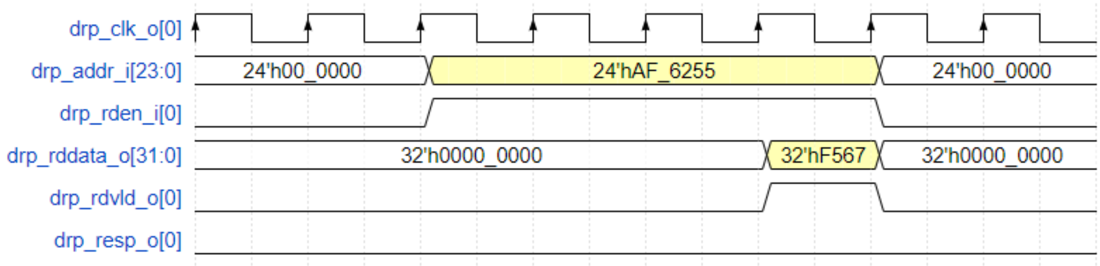
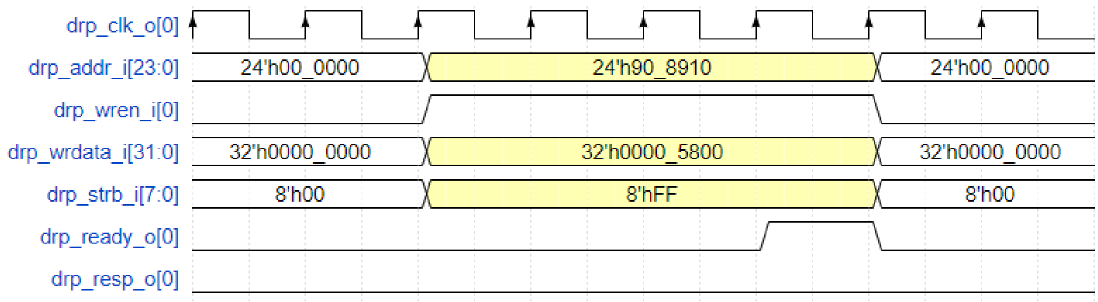


Figure 3-19 DRP Interface Write Timing Diagram



3.11.1 Clock Schematic Option

In Gowin SerDes, each Quad has two REFMUXs, including REFMUX0 and REFMUX1. REFMUX can receive input clocks from the reference clock pins of the current Quad or from the reference clock pins of adjacent Quad. REFMUX provides reference clocks for each TX PLL and RX CDR. In applications, users can select the reference clock source and PLL on the interface, and the software automatically calculates the MUX path to complete the clock connection.

Users can view the current clock connection through "View Clock Schematic" function on the SerDes interface, as shown in Figure 3-20.

Figure 3-20 Clock Schematic

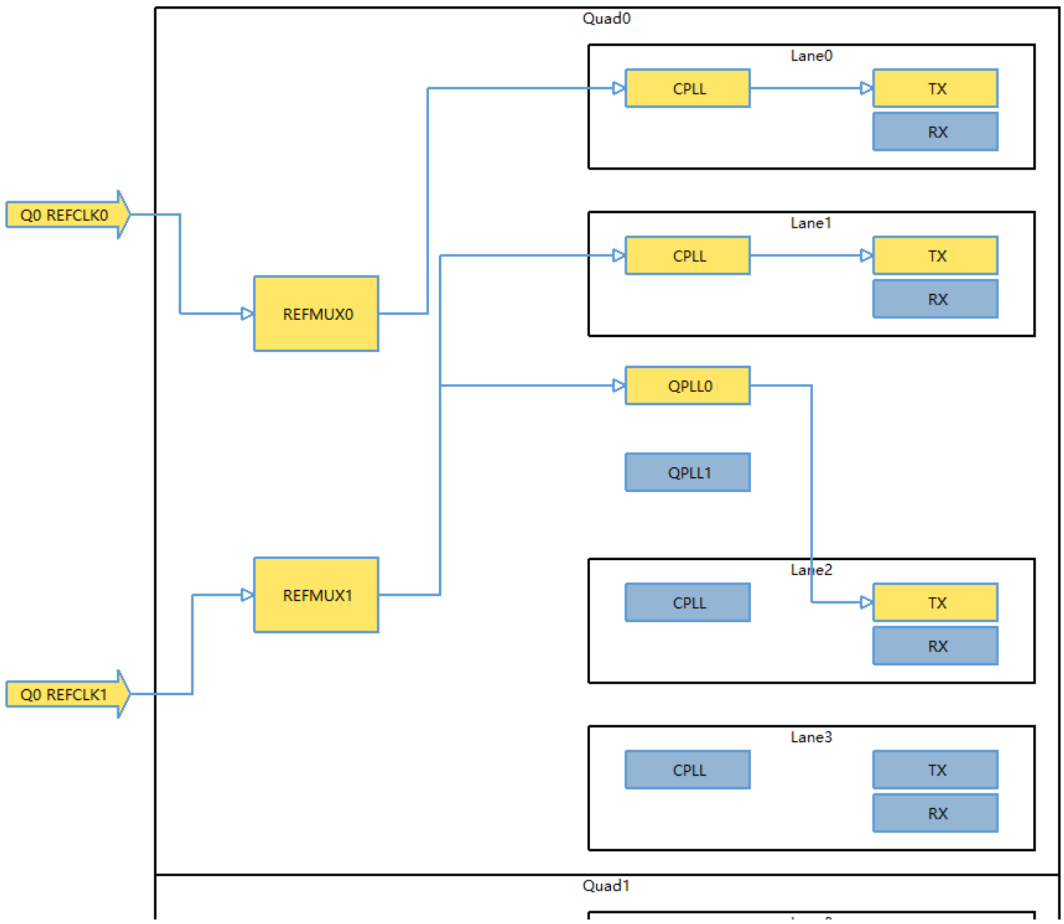


Figure 3-20 shows the clock connection with some of the functions having been configured. Occupied resources are highlighted in yellow, while unoccupied resources are highlighted in blue.

Taking Quad0 Lane0 as an example. This Lane TX uses CPLL as the TX clock source. The reference clock of CPLL comes from REFMUX0, and for REFMUX0, Q0 REFCLK0 is selected as the clock input. The reference clock source of the RX lane is selected automatically, which is not indicated in the above figure for now.

3.11.2 Reconfiguration Option

For GW5AT-138 devices, users can click the "Reconfiguration" button on Gowin SerDes IP interface to reconfigure, select the dynamic configuration functions and enter the parameters. Finally, click "Export" to export the register addresses and value file (.csr) required for configuring the current dynamic functions.

.csr File

The .csr file is a file containing dynamic configuration register addresses and values exported by the IP. Each line represents a register address + value, in hexadecimal format. The high 24 bits represent the register address, and the low 32 bits represent the register value.

For example, a line, such as 80a00400022322, indicates that 0x80a004 is the register address, and 0x00022322 is the value to be configured at that address. Users need to write the configurations from the .csr file to the SerDes via the DRP interface in the order from top to bottom, in order to achieve the dynamic configuration of the functions.

Channel Option

Check the channel that need to be dynamically configured this time.

Reconfiguration Option

This option includes different functions, such as TX Data Rate, RX Data Rate, etc. Under each function, there is an Enable option. If the user checks the Enable option for a certain function, the exported .csr file will include the dynamic configuration registers for this function. If the user does not check the Enable option for a certain function, the exported .csr file will not include the dynamic configuration registers for this function.

The descriptions of TX Data Rate are shown below.

- TX Data Rate: Used to configure the TX data rate
- TX PLL: Used to select the current TX PLL
- Refclk MUX: Used to select the PLL reference clock source as REFMUX0 or REFMUX1.
- Q0/1 REFMUX0/1 Frequency: Used to input the frequency of the REFMUX reference clock. Only input the frequency of the selected reference clock in the Refclk MUX option; for irrelevant reference clocks, it can be input as 0.
- PMA Width: Used to select the current PMA width

The descriptions of RX Data Rate are shown below.

- RX Data Rate: Used to configure the RX data rate
- Refclk MUX: Used to select the CDR reference clock source as REFMUX0 or REFMUX1.
- Q0/1 REFMUX0/1 Frequency: Used to input the frequency of the REFMUX reference clock. Only input the frequency of the selected reference clock in the Refclk MUX option; for irrelevant reference clocks, it can be input as 0.

The description of Loopback is shown below.

- Mode: Used to select loopback mode.

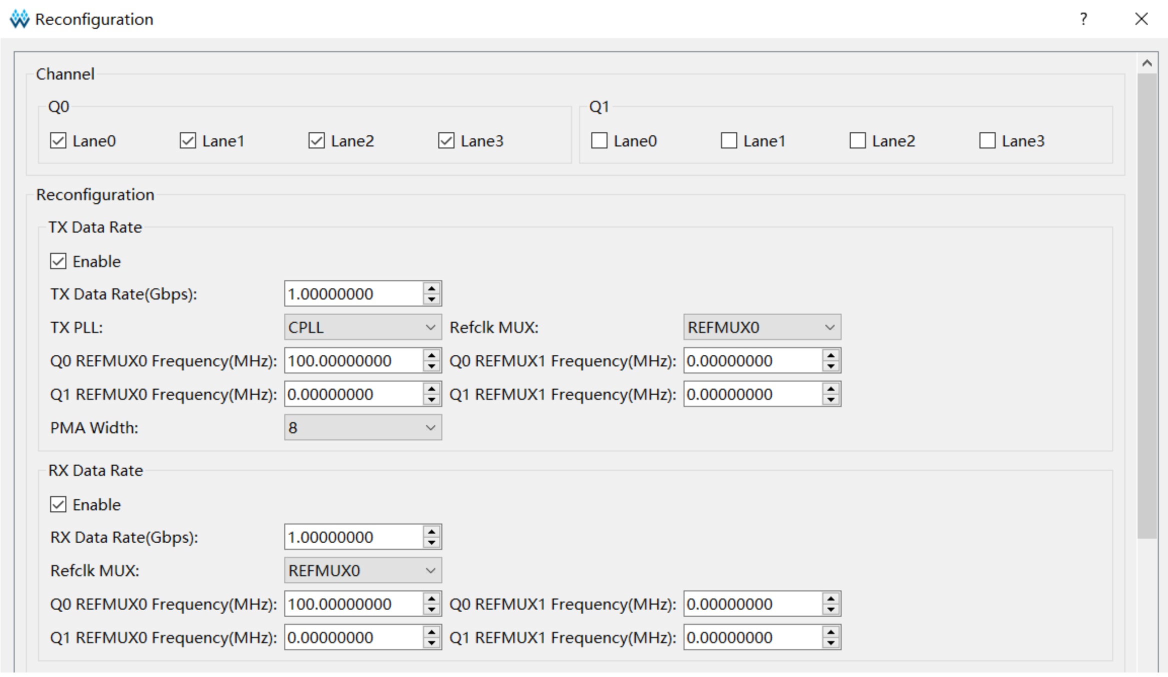
The descriptions of TX AFE are shown below.

- Mode: Used to select Auto or Manual.
- TX Swing Level: Used to configure TX Swing.
- FFE CM: Used to configure CM parameter, and it is valid when Manual mode is selected.
- FFE C1: Used to configure C1 parameter, and it is valid when Manual mode is selected.

For example, the dynamic configuration is as shown in Figure 3-21.

- The TX and RX data rate of Q0 Lane0/1/2/3 is 1Gbps.
- All reference clocks come from REFMUX0.
- The frequency of the reference clocks is 100MHz.
- CPLL is used to provide reference clock for TX.
- PMA Width is set to 8.

Figure 3-21 Reconfiguration Option



4 Port List

The detailed port list of Gowin Customized PHY IP is shown in Table 4-1; taking Quad0 Lane0 as an example, for other Lanes, you can modify the serial number.

Table 4-1 I/O List of Gowin Customized PHY IP

Port Name	I/O	Data Width	Description
Clock			
q0_ln0_rx_pcs_clkout_o	output	1	Quad0 Lane0 PCS RX clock output
q0_ln0_rx_clk_i	input	1	Quad0 Lane0 RX Buffer read clock, which can be directly connected to q0_ln0_rx_pcs_clkout_o, so that the RX Buffer read and write clocks are in the same frequency.
q0_ln0_tx_pcs_clkout_o	output	1	Quad0 Lane0 PCS TX clock output
q0_ln0_tx_clk_i	input	1	Quad0 Lane0 TX Buffer write clock, which can be directly connected to q0_ln0_tx_pcs_clkout_o, so that the TX Buffer read and write clocks are in the same frequency.
q0_ln0_cc_clk_i	input	1	When User is selected for Clock Source option on RX Clock Correction page, the user needs to input the clock through this port as the destination clock source for the adjustment clock of RX Clock Correction module.
q0_ln0_refclk_o	output	1	The divided output of the reference clock used by Quad0 Lane0
gpio_refclk0_i	input	1	GW5AT-60 GPIO reference clock input
gpio_refclk1_i	input	1	GW5AT-60 GPIO reference clock input
gpio_refclk_i	input	1	GW5AT-15 GPIO reference clock input
mclk_i	input	1	GW5AT-15 OSC reference clock input
Reset			
q0_ln0_pma_rstn_i	input	1	PMA reset input, active-low.
q0_ln0_pcs_rx_rst_i	input	1	PCS reset input in RX direction, active-high.
q0_ln0_pcs_tx_rst_i	input	1	PCS reset input in TX direction, active-high.

Port Name	I/O	Data Width	Description
RX Data Interface			
q0_ln0_rx_data_o	output	88	Quad0 Lane0 RX Buffer read data, synchronized with q0_ln0_rx_clk_i , valid when q0_ln0_rx_valid_o is 1.
q0_ln0_rx_fifo_rden_i	input	1	Quad0 Lane0 read enable, synchronized with q0_ln0_rx_clk_i. 1: Read valid 0: Read invalid
q0_ln0_rx_fifo_rdusewd_o	output	5	Remaining data indicator for Quad0 Lane0 RX Buffer
q0_ln0_rx_fifo_aempty_o	output	1	Quad0 Lane0 RX Buffer almost empty indicator 1: RX Buffer almost empty 0: RX Buffer non-almost empty
q0_ln0_rx_fifo_empty_o	output	1	Quad0 Lane0 RX Buffer empty indicator 1: RX Buffer empty 0: RX Buffer non-empty
q0_ln0_rx_valid_o	output	1	Quad0 Lane0 RX Buffer read data valid indicator, indicating whether q0_ln0_rx_data_o is valid. 1: Valid 0: Invalid
q0_ln0_rxc_o	output	8	Quad0 Lane0 64B66B RXC
q0_ln0_rxd_o	output	64	Quad0 Lane0 64B66B RXD
q0_ln0_rx_header_o	output	2	Quad0 Lane0 64B67B receives header
q0_ln0_rx_data_o	output	64	Quad0 Lane0 64B67B receives data
TX Data Interface			
q0_ln0_tx_data_i	input	80	Quad0 Lane0 TX Buffer write data, synchronized with q0_ln0_tx_clk_i; when q0_ln0_tx_fifo_wren_i is 1, write TX Buffer.
q0_ln0_tx_fifo_wren_i	input	1	Quad0 Lane0 write enable, synchronized with q0_ln0_tx_clk_i. 1: Write valid 0: Write invalid
q0_ln0_tx_fifo_wrusewd_o	output	5	Remaining data indicator for Quad0 Lane0 TX Buffer
q0_ln0_tx_fifo_afull_o	output	1	Quad0 Lane0 TX Buffer almost full indicator 1: TX Buffer almost full 0: TX Buffer non-almost full
q0_ln0_tx_fifo_full_o	output	1	Quad0 Lane0 TX Buffer full indicator 1: TX Buffer full 0: TX Buffer non-full
q0_ln0_tx_fetch_o	output	1	Indication for fetching Quad0 Lane0 64B66B/64B67B TX data 1: TX data fetched 0: TX data not fetched
q0_ln0_txc_i	input	8	Quad0 Lane0 64B66B TXC

Port Name	I/O	Data Width	Description
q0_ln0_txd_i	input	8	Quad0 Lane0 64B66B TXD
Control Interface			
q0_ln0_cb_start_i	input	1	This port is active when IP enables RX Channel Bonding. When the user pulls up this pin, the Channel Bonding module inside SerDes starts aligning data. The user needs to wait until the word_align_link_o of the selected channels are all 1 before simultaneously pulling up this pin of the selected channel.
q0_ln0_tx_ctrl_i	input	3	Quad0 Lane0 64B66B/64B67B TX control signal, and need to be assigned 3'b000.
Status Interface			
q0_ln0_signal_detect_o	output	1	RX differential signal status indicator 1: Valid signal input is detected 0: No valid signal input is detected and RX is in Electrical Idle.
q0_ln0_rx_cdr_lock_o	output	1	RX CDR lock indicator 1: CDR locked 0: CDR unlocked
q0_ln0_k_lock_o	output	1	Pre-lock indicator for RX word align module 1: Word align module in pre-lock status 0: Word align module not in pre-locked status
q0_ln0_word_align_link_o	output	1	Lock indicator for RX word align module 1: Word align module in lock status 0: Word align module not in lock status
q0_ln0_pll_lock_o	output	1	TX PLL lock indicator 1: TX PLL locked 0: TX PLL unlocked
q0_ln0_ready_o	output	1	TX channel status indicator 1: Ready 0: Not ready
q0_ln0_tx_invld_blk_o	output	1	Quad0 Lane0 64B66B/64B67B receives invalid block 1: invalid block received 0: invalid block not received
q0_ln0_rx_blk_lock_o	output	1	Indication for Quad0 Lane0 64B66B/64B67B protocol block lock 1: Locked 0: Not locked
q0_ln0_rx_dec_err_o	output	1	Quad0 Lane0 64B66B/64B67B protocol decoder error indication 1: Decode error 0: Decode correct
q0_ln0_rx_dscr_err_o	output	1	Quad0 Lane0 64B67B descrambler error indication 1: descrambler error 0: descrambler correct

Port Name	I/O	Data Width	Description
q0_ln0_rx_invld_header_o	output	1	Quad0 Lane0 64B66B/64B67B protocol invalid sync header indication 1: invalid sync header 0: valid sync header
q0_ln0_rx_ctc_ins_o	output	2	Indication for Quad0 Lane0 64B66B CTC module inserting word bit[1]: 1: upper word (higher four-byte) is added word bit[0]: 1: lower word (lower four-byte) is added word
q0_ln0_rx_ctc_del_o	output	2	Indication for Quad0 Lane0 64B66B CTC module deleting word bit[1]:1: the previous upper word (higher four-byte) is deleted bit[0]: 1: the previous lower word (lower four-byte) is deleted
q0_ln0_rx_her_hi_o	output	1	Quad0 Lane0 64B66B/64B67B protocol high BER indication 1: high BER
Dynamic Configuration Interface			
drp_clk_o	output	1	DRP interface clock
drp_addr_i	input	24	DRP operation address, synchronized with drp_clk_o.
drp_wren_i	input	1	DRP write operation enable, synchronized with drp_clk_o. 1: Write operation 0: No operation
drp_wrdata_i	input	32	DRP write data, synchronized with drp_clk_o.
drp_strb_i	input	8	DRP write operation enable signal, active-high, synchronized with drp_clk_o. All bits of this signal need to be set to 1 during write operation.
drp_ready_o	output	1	DRP write operation completion indicator, synchronized with drp_clk_o. 1: Completed 0: Not completed
drp_rden_i	input	1	DRP read operation enable, synchronized with drp_clk_o. 1: Read operation 0: No operation
drp_rdvld_o	output	1	DRP read operation data valid indicator, synchronized with drp_clk_o 1: Returned data valid 0: No valid data returned
drp_rddata_o	output	32	DRP read data, synchronized with drp_clk_o.
drp_resp_o	output	1	Reserved

5 Interface Configuration

You can call and configure Gowin Customized IP using the IP core generator tool in the IDE.

1. Open SerDes IP

After creating the project, click the "Tools" tab in the upper left interface, click "IP Core Generator" from the drop-down list to open Gowin IP Core Generator, then double-click to open "Serdes IP".

2. Open Customized IP Core

After opening SerDes IP, find "Customized" in the "Protocol" drop-down list and click "Create" to open Customized IP protocol configuration interface.

3. Configure Customized IP

The Customized IP configuration interface is shown in Figure 5-1, Figure 5-2，Figure 5-3, Figure 5-4，Figure 5-5, Figure 5-6, and Figure 5-7, including "Channel, Line Rate, Refclk Selection", "Data Width, Encoding, RX Word Alignment", "Channel Bonding", "RX Clock Correction" and "AFE". You can configure Customized IP parameters on these option pages. See Table 5-1 for the descriptions of the parameters.

Figure 5-1 Channel, Line Rate, Refclk Selection for GW5AT-138

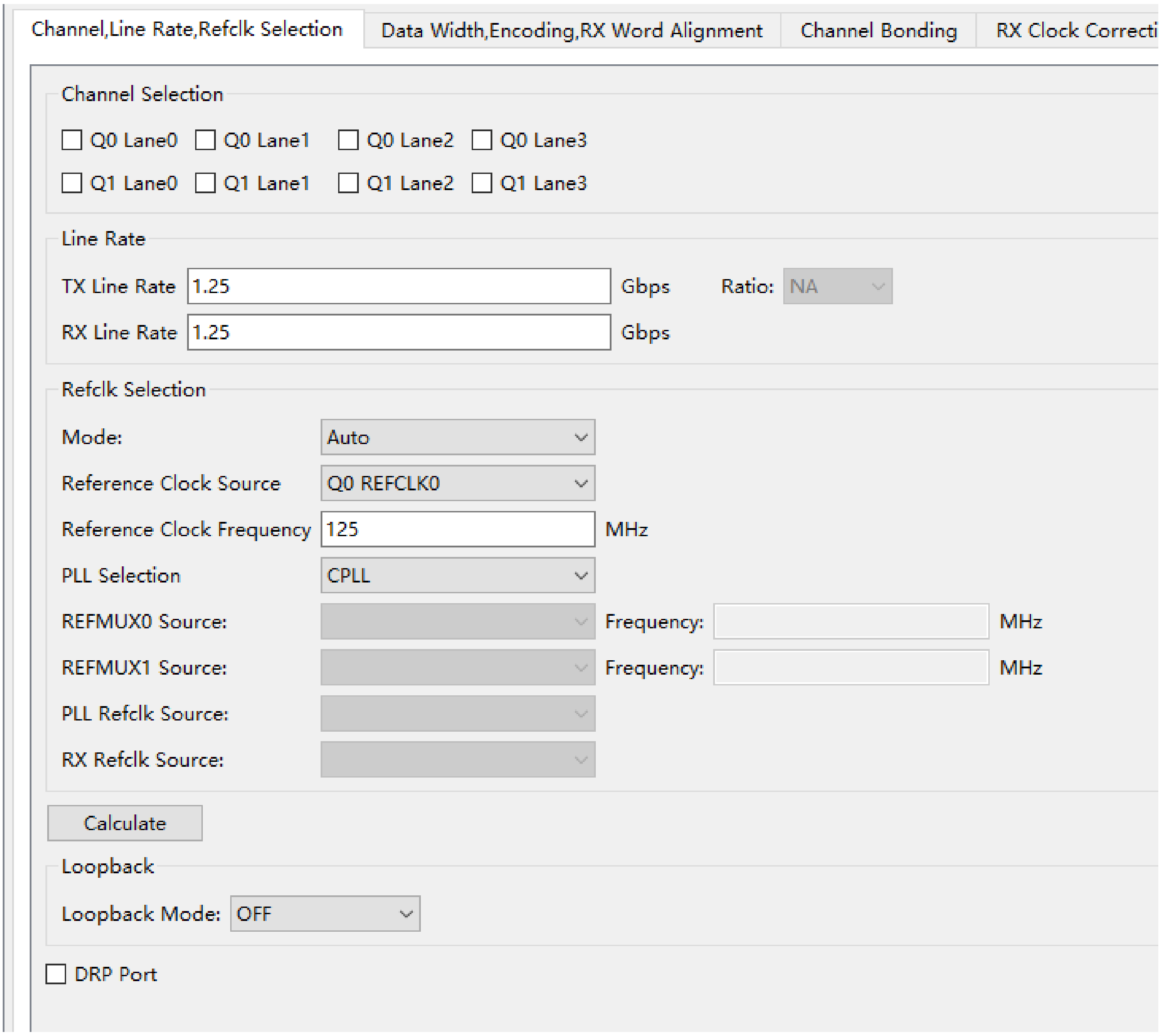


Figure 5-2 Channel, Line Rate, Refclk Selection for GW5AT-60

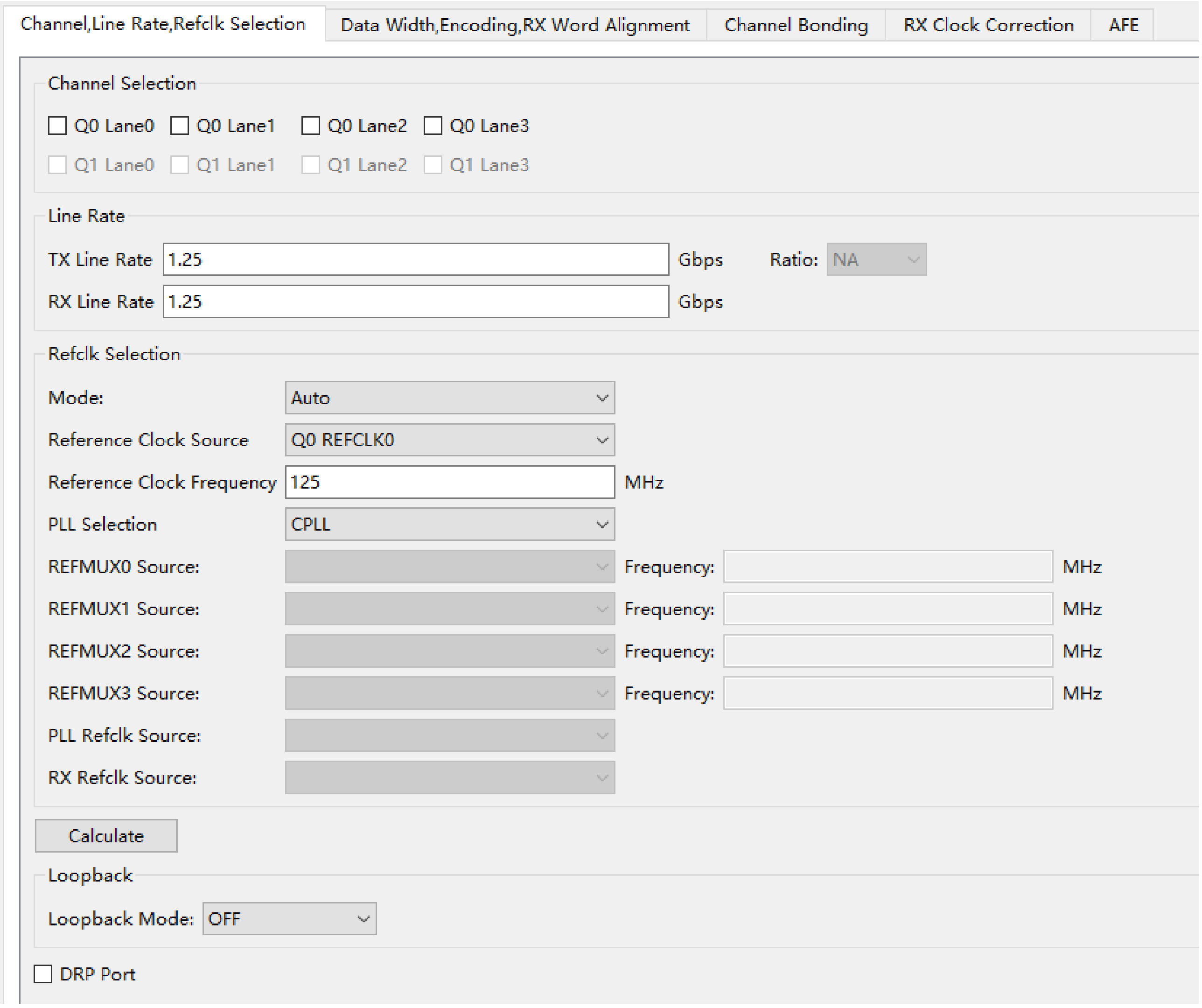


Figure 5-3 Data Width, Encoding, RX Word Alignment for GW5AT-138

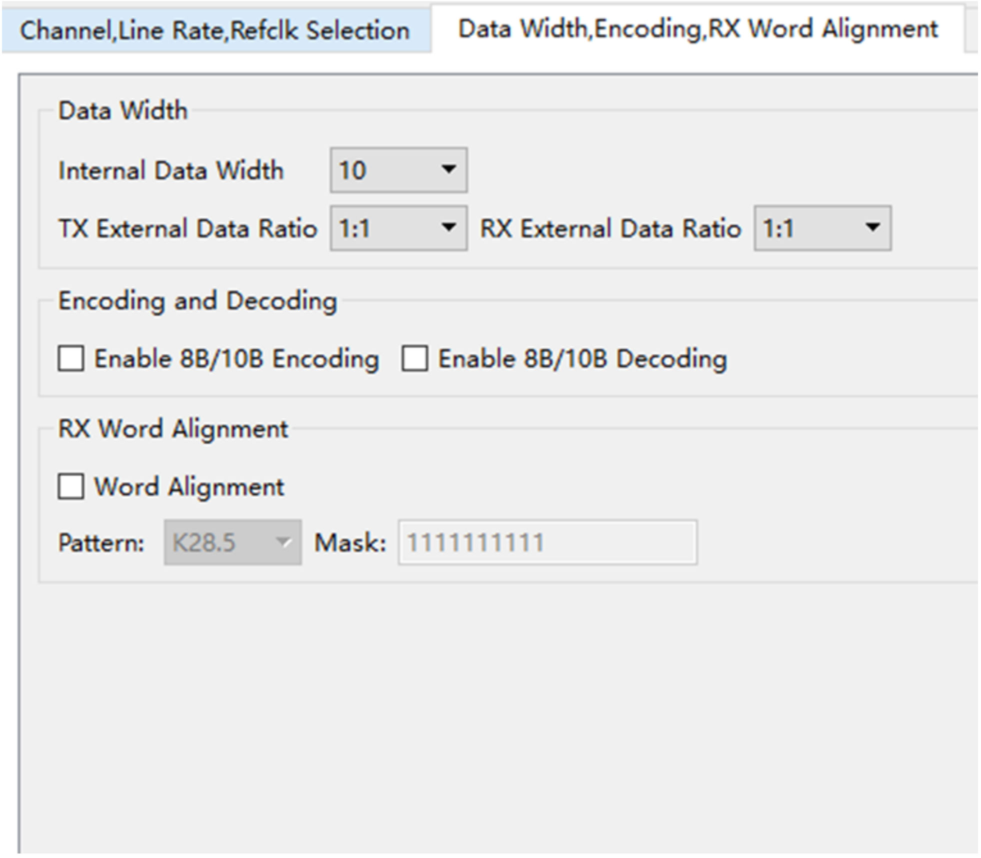


Figure 5-4 Data Width, Encoding, RX Word Alignment for GW5AT-60

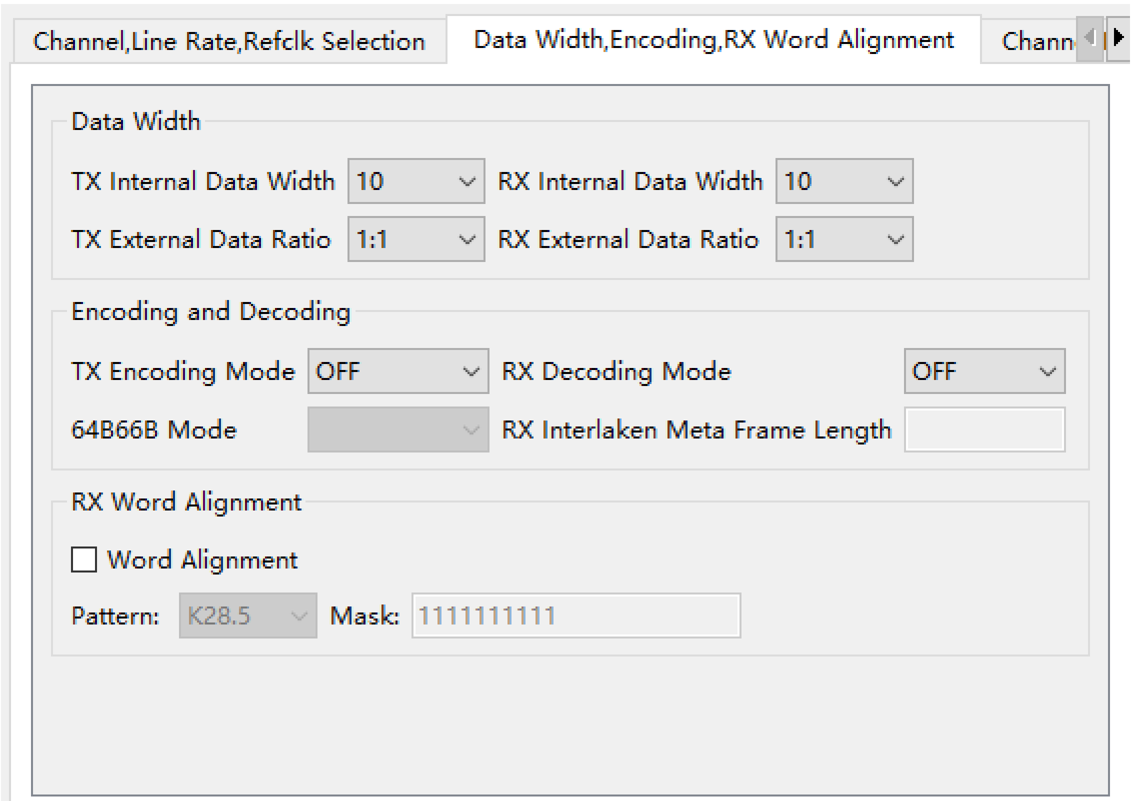


Figure 5-5 Channel Bonding

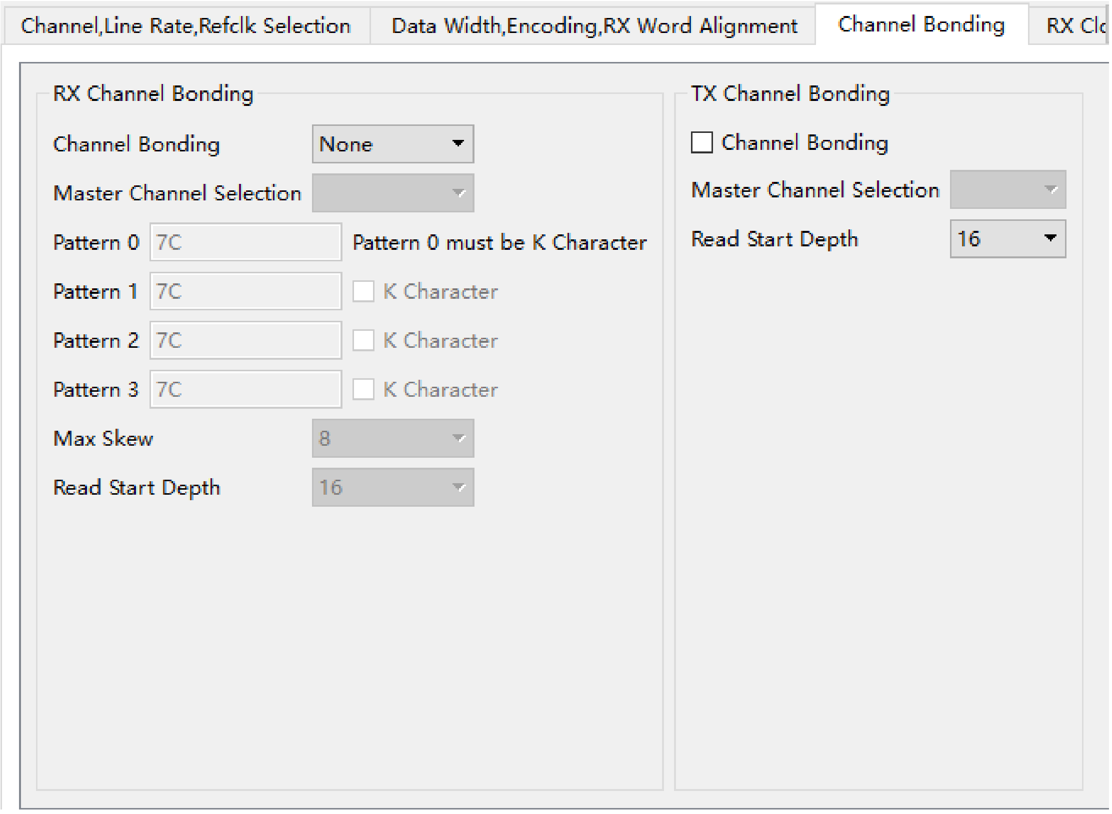


Figure 5-6 RX Clock Correction

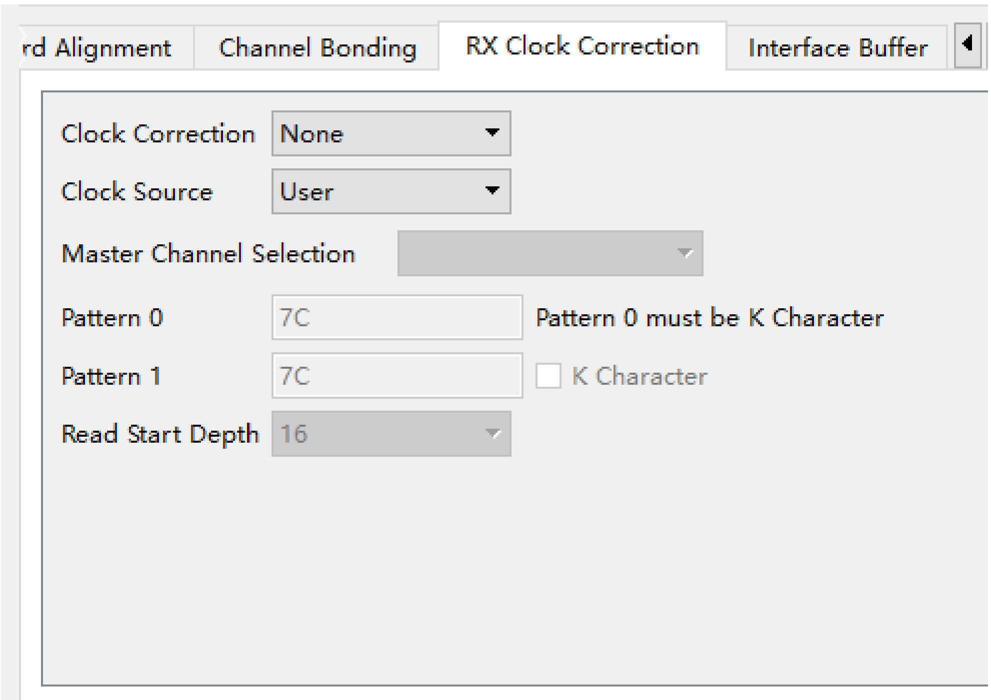
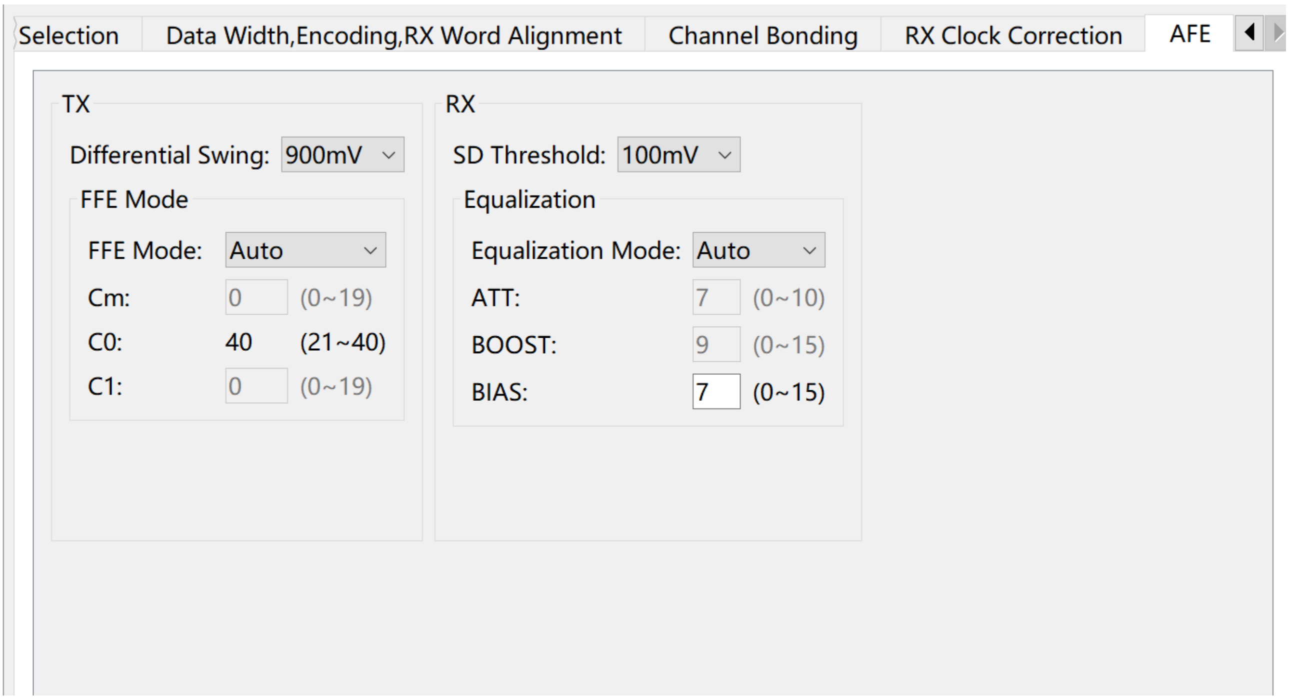


Figure 5-7 AFE



After configuring Customized IP parameters, click the "OK" button to generate the configuration associated with Customized IP.

4. Complete SerDes IP configuration

On the SerDes IP interface, the user completes the configuration of all protocols and then clicks "OK" button to complete the generation of the SerDes IP. In the SerDes IP top-level file, the signals used by this IP have the same prefix as the Module Name of the Customized PHY IP interface.

Table 5-1 Customized IP Parameters

Name	Range	Description
Channel, Line Rate, Refclk Selection		
Channel Selection	Q0 Lane0 Q0 Lane1 Q0 Lane2 Q0 Lane3 Q1 Lane0 Q1 Lane1 Q1 Lane2 Q1 Lane3	The user can check any one or more Lanes; If the user checks one Lane, all subsequent configurations are for this Lane; if the user checks more Lanes, all subsequent configurations are for all the checked Lanes.
Operation Mode	TX and RX TX Only RX Only	Operation mode selection TX and RX: The selected lane is used as a bidirectional data lane. TX Only: The selected lane is used only as a transmit data lane. RX Only: The selected lane is used only as a receive data lane.
TX Line Rate	0.025Gbps~12.5Gbps	Configure transmit data rate

Name	Range	Description
Ratio	NA 5x 10x 20x 40x	Configure the TX data rate ratio when TX Line Rate < 1Gbps. See TX Line Rate < 1Gbps for details
RX Line Rate	1Gbps~12.5Gbps	Configure receive data rate
Mode	Auto Manual	Clock mode selection Auto: Automatic mode Manual: Manual mode If there are no special requirements, it is recommended to choose Auto.
Reference Clock Source	Q0 REFCLK0 Q0 REFCLK1 Q1 REFCLK0 Q1 REFCLK1 Q0 REFIN0 Q0 REFIN1 Q0 REFIN MCLK	Select reference clock source The reference clock sources vary depending on the chip. For details, you can see 3.2 Reference Clock.
Reference Clock Frequency	50~800MHz	Configure reference clock source
PLL Selection	CPLL QPLL0 QPLL1	Select PLL **Note! CPLL is recommended for a single lane; QPLL is required for multiple lanes.**
REFMUX0 Source	Q0 REFCLK0 Q0 REFCLK1 Q1 REFCLK0 Q1 REFCLK1 Q0 REFIN0 Q0 REFIN1	Select the reference clock source for REFMUX0. The reference clock sources vary depending on the chip. For details, you can see 3.2 Reference Clock.
REFMUX1 Source	Q0 REFCLK0 Q0 REFCLK1 Q1 REFCLK0 Q1 REFCLK1 Q0 REFIN0 Q0 REFIN1	Select the reference clock source for REFMUX1. The reference clock sources vary depending on the chip. For details, you can see 3.2 Reference Clock.
REFMUX2 Source	Q0 REFCLK0 Q0 REFCLK1 Q1 REFCLK0 Q1 REFCLK1 Q0 REFIN0 Q0 REFIN1	Select the reference clock source for REFMUX2. The reference clock sources vary depending on the chip. For details, you can see 3.2 Reference Clock.
REFMUX3 Source	Q0 REFCLK0 Q0 REFCLK1 Q1 REFCLK0 Q1 REFCLK1 Q0 REFIN0	Select the reference clock source for REFMUX3. The reference clock sources vary depending on the chip. For details, you can see 3.2 Reference Clock.

Name	Range	Description
	Q0 REFIN1	
PLL Refclk Source	REFMUX0 REFMUX1 REFMUX2 REFMUX3 GPIO MCLK	Select the PLL reference clock source. The reference clock sources vary depending on the chip. For details, you can see 3.2 Reference Clock.
RX Refclk Source	REFMUX0 REFMUX1 REFMUX2 REFMUX3	Select the receive reference clock source. The reference clock sources vary depending on the chip. For details, you can see 3.2 Reference Clock.
Loopback Mode	OFF LB_NES LB_FES LB_ENC	OFF: No Loopback, normal operating mode LB_NES: Loopback Near-End Side LB_FES: Loopback Far-End Side LB_ENC: Loopback Encoder
Calculate	-	Check if the reference clock frequency matches the data rate; if it matches, "Succeed" will pop up.
DRP Port	Checked/Unchecked	Checked: Enable DRP Unchecked: Disable DRP
Data Width, Encoding, RX Word Alignment		
Internal Data Width	8 10 16 20	SerDes internal data bit width
TX External Data Ratio	1:1 1:2 1:4	TX interface data width ratio TX interface data width ratio = Internal Data Width* TX External Data Ratio
RX External Data Ratio	1:1 1:2 1:4	RX interface data width ratio RX interface data width ratio = Internal Data Width* RX External Data Ratio
Enable 8B/10B Encoding	Checked/Unchecked	Checked: Enable TX 8B/10B encoding Unchecked: Disable TX 8B/10B encoding
Enable 8B/10B Decoding	Checked/Unchecked	Checked: Enable RX 8B/10B decoding Unchecked: Disable RX 8B/10B decoding
TX Internal Data Width	8 10 16 20	SerDes TX internal data width
RX Internal Data Width	8 10 16 20	SerDes RX internal data width
TX Encoding Mode	OFF 8B10B 64B66B	TX data encoding format

Name	Range	Description
	64B67B	
RX Decoding Mode	OFF 8B10B 64B66B 64B67B	RX data decoding format
64B66B Mode	10GBASER with FIFO Async with FIFO	64B66B mode selection
Interlaken Meta Frame Length	5~16383	64B67B meta frame length in word (64bits)
Word Alignment	Checked/Unchecked	Checked: Enable RX Word Alignment Unchecked: Disable RX Word Alignment
Pattern	Valid K character such as K28.0, K28.5, etc.	Select Word Alignment type
Mask	0000000000~ 1111111111	Pattern mask; when word aligning, comparing bits where the mask is 1 and not comparing bits where the mask is 0
RX Channel Bonding		
Channel Bonding	None One Word Two Words Four Words	RX Channel Bonding enable: None: Disable Channel Bonding One Word: Enable Channel Bonding for 1 Word Two Words: Enable Channel Bonding for 2 Words Four Words: Enable Channel Bonding for 4 Words
Master Channel Selection	Lane selected	Select master channel for RX Channel Bonding
Pattern0	0x00~0xFF	The first alignment character must be K character
Pattern1/2/3	0x00~0xFF	For the 2nd/3rd/4th alignment character, it can be K character or not
K Character	Checked Unchecked	Configure whether the alignment character is K character or not Checked: K character Unchecked: Data
Max Skew	0~31	Configure the maximum skew for inter-RX channels
Read Start Depth	0~31	After bonding the data, configure the read start depth of the module
Channel Bonding		
Channel Bonding	Checked/Unchecked	TX Channel Bonding enable Checked: Enable TX Channel Bonding Unchecked: Disable TX Channel Bonding
Master Channel Selection	Lane selected	Select master channel for TX Channel Bonding
Read Start Depth	0~31	After bonding the data, configure the read start depth of TX Buffer
RX Clock Correction		

Name	Range	Description
Clock Correction	None One Word Two Words	RX Clock Correction enable: None: Disable Clock Correction One Word: Enable Clock Correction for 1 Word Two Words: Enable Clock Correction for 2 Words
Clock Source	User Quad TX	Select the destination clock source that needs to be synchronized with RX Clock Correction User: Set Clock Source as Fabric input clock (e.g. q0_ln0_cc_clk_i) TX: Set the Clock Source to TX clock Quad: Reserved
Master Channel Selection	Lane selected	Select master channel for RX Clock Correction
Pattern0	0x00~0xFF	The first adjustment character must be K character
Pattern1	0x00~0xFF	For the 2nd adjustment character, it can be K character or not
K Character	Checked Unchecked	Configure whether the adjustment character is K character or not Checked: K character Unchecked: Data
Read Start Depth	0~31	After adjusting data, configure the read start depth of the module
AFE		
Differential Swing	100mV~900mV	Configure TX differential signal swing Vdiffpp, Vdiffpp=2xVdiff
FFE Mode	Auto Manual	Configure TX FFE mode ● Auto: Automatic mode ● Manual: Manual mode
Cm	0~19	Transmit FFE pre-cursor
C0	21~40	Transmit FFE main-cursor
C1	0~19	Transmit FFE post-cursor
SD Threshold	25mV~200mV	Receive differential signal SD threshold
Equalization Mode	Auto Manual	Receive Equalization: ● Auto: Automatic mode ● Manual: Manual mode
ATT	0~10	Adjust the intermediate frequency attenuation during reception. A smaller value indicates greater attenuation.
BOOST	0~15	Adjust the high-frequency amplification during reception. A larger value indicates greater gain.
BIAS	0~15	Configure the amplification parameters of the SerDes for RX signals. A higher value indicates stronger signal amplification.

6 Reference Design

See the Customized PHY reference design for details at Gowinsemi website.